

Informed Trading in Foreign Exchange Futures: Payroll News Timing

Yang-Ho Park

Federal Reserve Board

December 12, 2021

ABSTRACT

This paper studies informed trading about U.S. payrolls in the foreign exchange (FX) futures market. I find that speculators such as hedge funds are more likely to be sellers than buyers of FX futures ahead of good U.S. payroll news and thus appear to have earned significant gains around payroll announcements. In contrast, hedgers—in particular, dealers—appear to have provided liquidity to speculators. I show that mimicking speculators' FX exposures around payroll announcements can add a large economic gain to various reference portfolios. My analysis also uncovers that information in FX trading is long-lived and differs along the U.S. business cycle.

JEL classification: F31; G13; G15

Keywords: foreign exchange; informed trading; macroeconomic announcements; futures market; U.S. dollar

I am grateful for comments and suggestions from Carol Alexander (editor), Alain Chaboud, Andreas Rapp, and Rui Albuquerque (associate editor). Disclaimer: The analysis and conclusions set forth are those of the author and do not indicate concurrence by the Board of Governors of the Federal Reserve System or other members of its research staff. Send correspondence to Yang-Ho Park, Risk Analysis Section, Board of Governors of the Federal Reserve System, 20th & C Streets, NW, Washington, D.C., 20551, Tel: (202) 452-3177, e-mail: yang-ho.park@frb.gov.

1 Introduction

Asymmetric information and trading play a crucial role in understanding the workings of the FX markets. A leading example with a microstructure perspective is the tight relationship between FX rates and order flows, where order flows are a measure of net buying pressure or an aggregator of heterogeneous information (see Evans and Lyons 2002). Subsequent authors have made efforts to clarify economic origins of the relationship because order flows can be driven by not only macroeconomic information but also liquidity and hedging demands.¹ There is also a large literature showing that order flows have predictive information for FX returns.² Using the data on customers' FX trades, some authors have attempted to identify informed traders and explain FX rate fluctuations through the lens of their trading.³

Despite the abundance of micro-based FX research, it predominantly concentrates on over-the-counter (OTC) contracts, with little focus on futures contracts. On the one hand, such predominance is understandable because the OTC market combined is one order of magnitude larger than the futures market and because some proprietary OTC data allow identification of certain traders' order flows. On the other hand, little research on the futures market is surprising because futures have some distinctive features that can attract informed traders. For example, all-to-all anonymous trading in futures allows informed traders to hide their private information, which would be more difficult on banks' trading platforms. To the extent that information is long-lived, informed traders may prefer limit orders in the futures market to earn a bid-ask spread.⁴ In addition, the futures market is transparent (e.g., has a common price), integrated, and standardized, thereby providing a level playing field, unlike

¹ See Bacchetta and Van Wincoop (2006), Evans and Lyons (2008), Love and Payne (2008), and Berger et al. (2008), among many others.

² See, for example, Evans and Lyons (2005), Rime et al. (2010), King et al. (2010), and Menkhoff et al. (2016).

³ See, for example, Sager and Taylor (2008), Cerrato et al. (2011), and Menkhoff et al. (2016).

⁴ Kaniel and Liu (2006) offer an equilibrium trading model in which informed traders prefer limit orders to market orders and limit orders are more informative. Collin-Dufresne and Fos (2015) and Kacperczyk and Pagnotta (2019) provide empirical evidence that informed traders use limit orders.

the OTC market. Finally, FX futures have no credit valuation adjustments unlike OTC derivatives.⁵

This paper aims to study informed trading and liquidity provision in FX futures with respect to the U.S. payroll announcements. Arguably, traders consider payroll news to be most important in FX trading, often calling it the “king” of announcements (Andersen and Bollerslev 1998).⁶ The analysis is facilitated by investors’ FX exposures that I derive from open interest snapshots published by the Commodity Futures Trading Commission (CFTC). Using the exposure measures on the seven largest FX futures, I provide evidence of informed trading in FX futures, with an emphasis on different roles played by speculators and hedgers. In addition, I introduce a payroll news timing strategy conditioned on investors’ FX exposures to boost the performance of portfolios of any kind.

A chief finding is that the pre-payroll-news FX exposures—the FX exposures that each investor type has taken ahead of payroll announcements—are associated with the subsequent payroll news. Specifically, speculators have rebalanced their FX exposures ahead of payroll announcements in the right direction rather than in the wrong direction. In stark contrast, hedgers have held opposite positions from speculators, adjusting their FX exposures in the wrong direction rather than in the right direction. The pre-payroll-news divergence between speculators and hedgers is most pronounced for the euro, Japanese yen, and Australian dollar. In light of this finding, speculators appear to be informed traders in the futures market, whereas hedgers act like liquidity providers.

To highlight the different roles of speculators and hedgers, I estimate investors’ FX portfolio earnings—the returns that each investor type might have earned on their FX futures portfolio around payroll announcements under some assumptions. A striking pattern emerges from these estimates. As can be seen in Figure 1, speculators’ portfolio earnings jump

⁵ In addition, for the firms subject to the Basel III leverage ratio requirement or the supplementary leverage ratio rule in its U.S. implementation, futures can offer regulatory capital savings as they are centrally cleared.

⁶ The employment status constitutes the dual mandate of the Federal Reserve and Gilbert et al. (2017) find that, among various macro news, payroll news is most important in the dynamics of Treasury yields.

immediately after the 8:30 a.m. payroll announcements (the top panel), whereas hedgers' portfolio earnings plunge immediately after the payroll announcements (the middle panel). These results also imply that speculators are informed traders with respect to payroll news and hedgers are liquidity providers.

I evaluate the economic value of information in investors' FX exposures by introducing a payroll news timing strategy. The timing strategy starts with an assumed reference portfolio comprising risky assets and cash holdings and attempts to augment the portfolio performance by trading an FX portfolio just before a payroll announcement and unwinding it any time before the day's market close. The key idea of the timing strategy lies in constructing a target FX portfolio by exactly mimicking informed traders' FX exposures. Note that the FX portfolio formation is model-free and requires only a single snapshot of the FX exposures immediately preceding a payroll release. As a result, the strategy is out-of-sample by nature and can be implemented very easily in real time.

My analysis shows that the timing strategy can deliver a significant economic gain over various reference portfolios, including stocks, bonds, and commodities. Mimicking speculators' exposures can easily deliver a Sharpe ratio gain of 0.3 to 0.4, regardless of reference portfolios, with some reasonable amount of cash available for the required margin. The robust improvement is due to the fact that returns of the timed FX portfolio have little correlations with monthly excess returns of any risky assets—the diversification effect. I also discuss the extent to which some realistic features of trading, such as transaction cost and smooth unwinding, impair the performance of the timing strategy.

I document a salient characteristic of informed trading in FX futures—information in FX exposures is long-lived. Specifically, current FX exposures have information about not only the nearest payroll announcement but also the second nearest one. I consider two explanations likely: timing- and stealth-based motives. First, it is possible that speculators are patient in executing their trades, probably, waiting for great noise trading activity and market liquidity. Related to this possibility, Collin-Dufresne and Fos (2015, 2016) show

that informed traders account for noise trading activity and market liquidity when executing trades. Authors such as Kacperczyk and Pagnotta (2019) and Ahern (2020) provide evidence of the strategic timings of informed trades. Such timings can slow down information percolation into futures prices. Second, it is also possible that speculators try to avoid revealing private information to other investors. Such stealth trading is particularly relevant when some speculators have exclusively private information that is difficult for other traders (and professional forecasters) to discern. For example, Foster and Viswanathan (1994) provide a strategic trading model in which stealth trading can delay the incorporation of exclusively private information into asset prices.

In addition to the long-lived nature, information in FX exposures depends on phases of the U.S. business cycle. Specifically, the payroll timing strategy is effective only during the National Bureau of Economic Research (NBER)-designated expansions but not effective during the recessions. This result is somewhat surprising because some existing research shows that economic agents process macroeconomic information more actively during recessions than during expansions.⁷ How can my finding be reconciled with the existing research? In fact, informed trading during recessions is complicated by the fact that the bad news of U.S. economic activity can cause a surprising appreciation of the U.S. dollar. In other words, even if some investors have superior payroll information during recessions, it would be difficult to take advantage of such information because safe-haven demands offset the bad-news effect on the U.S. dollar. I thus interpret my finding to suggest the difficulty of timing the FX market during recessions rather than to suggest traders' inability to process information in bad times.

A large body of FX research has studied the (un)predictability of FX rates since the pioneering work of Meese and Rogoff (1983). Having found evidence that FX exposures contain special information about future payrolls, it is natural to expect that FX exposures have predictive power for U.S. dollar fluctuations too, provided that payroll news is an important

⁷ See, for example, Kacperczyk et al. (2014), Coibion and Gorodnichenko (2015), and Park (2021).

driver of the U.S. dollar. Indeed, a common factor of FX exposures across different futures contains predictive information about U.S. dollar index returns in the following months. While the predictive power is achieved both in sample and out of sample, it is most evident when the common factor is aligned with the impending payroll surprise through a three-pass filter from Kelly and Pruitt (2015). I thus argue that information about future payrolls is a source of the predictability of FX exposures for U.S. dollar fluctuations.

In an attempt to understand who is truly driving information in FX exposures, I study the connection between speculators' FX exposures and hedge funds' stock exposures. By estimating a hedge fund return model in which the stock beta is tied to a common factor of FX exposures, I find that a one-standard-deviation shock in speculators' common factor is (correctly) associated with a 0.04–0.05 increase in aggregated hedge funds' stock beta. Furthermore, while the extent of the connection varies among different hedge fund styles, the strongest connection is found with hedge funds classified as managed futures. An explanation for the connection is that hedge funds' informed trading is a common driver of both the FX and stock market timings. Related to this explanation, Albuquerque et al. (2008) document the information connection between the stock and FX markets. Overall, I interpret the connection to imply that information in FX exposures originates from hedge funds, at least in part.

This paper should be distinguished from the existing literature using the futures positions data for the prediction of FX returns. For example, Wang (2004) finds that net futures positions contain predictive information for FX returns. Klitgaard and Weir (2004) find the contemporaneous relationship (but not the predictive relationship) between net futures positions and futures returns. Tornell and Yuan (2012) show that some measures of peaks and troughs in speculators' futures positions are predictive of spot FX returns. Although the predictive power of futures positions is well established in FX markets, the existing studies typically associate the predictability with market risk and hedge pressure effects in futures markets. That is, the existing literature says little about whether speculators have

superior macroeconomic information and, if so, what kind of information they have. As opposed to the literature, my paper provides evidence that speculators' payroll expectations are important in explaining variations in FX trading and prices.

My paper contributes to the literature studying asset returns on days of macroeconomic announcements or immediately before macroeconomic announcements. For example, Bernile et al. (2016) find that abnormal pre-FOMC order imbalances are aligned with subsequent policy surprises and attribute the alignment to information leakage. Kurov et al. (2019) discover pre-announcement drifts before several macroeconomic announcements. Savor and Wilson (2013) find that stock returns and Sharpe ratios are higher on days of major macroeconomic announcements. Mueller et al. (2017) find that the U.S. dollar tends to depreciate relative to other currencies on days of scheduled FOMC announcements. Ai and Bansal (2018) develop revealed preference theory for macroeconomic announcement premiums. Related to this literature, I show that payroll surprises are predictable using lagged information, suggesting that speculators' superior information may explain a fraction of payroll announcement returns.

Finally, this paper adds to the aforementioned literature that uses customers' trades data in the OTC market. There are similarities and differences between my futures-market study and the existing OTC-market studies. My result pointing to speculators as informed traders is consistent with the OTC-market result that financial clients are likely to be informed. For example, Cerrato et al. (2011) show that profit-seeking investors such as hedge funds and asset managers are informed traders, and Menkhoff et al. (2016) show that long-term asset managers are the most informed traders. In terms of the differences, the OTC-market research generally finds that non-financial corporations are liquidity providers, at least over short horizons, with some ambiguity about the role of dealers; dealers can be both liquidity providers and informed traders. However, my results suggest that hedgers—in particular, dealers—have been primary liquidity providers in the futures market.

The rest of the paper is organized as follows: Section 2 introduces the data and the

construction of key variables; Section 3 discusses informed trading and liquidity provision in the futures market; Section 4 introduces a payroll news timing strategy and evaluates its economic gains in various settings; Section 5 examines the information content of FX exposures for future U.S. dollar returns; Section 6 reports the connection between FX exposures and hedge funds' stock timing; and Section 7 concludes.

2 Data and variable construction

This section describes the data and key variables to study informed trading and liquidity provision in FX futures.

2.1 Data

FX exposures: FX exposures data are drawn from the CFTC's Commitments of Traders (COT) reports over the sample period from January 1988 through December 2018. I particularly utilize the futures-only data set which has a longer time series than the options-inclusive data set. The CFTC collects each Tuesday's snapshot of open interests submitted by futures commission merchants and publishes the data on the Friday of the same week.⁸ The total open interest is divided into reportable and non-reportable positions depending on whether reporting firms meet minimum threshold levels, which vary over time and across contracts. The reportable category is further divided into non-commercial and commercial categories based on the CFTC Form 40 in which traders identify their primary motive of trading. Non-commercial traders refer to those who identify themselves as using futures contracts for reasons other than hedging. In contrast, commercial traders refer to those who identify themselves as using futures contracts for hedging. It is conventional to refer to non-commercial and commercial traders as speculators and hedgers, respectively, and I henceforth follow the convention.

⁸ The data collection and publication took place fortnightly before October 1992.

I acknowledge a limitation of the COT data for the purpose of my study. Because of the time difference between the COT data reference time (the close of Tuesday) and the payroll announcement time (8:30 a.m., Friday), the COT data leave out information from Wednesday until the payroll announcement time. Thus, my analyses that follow rely on the informed trading that might have occurred on or before the Tuesday immediately preceding the payroll announcement. Early (or gradual) informed trading of such kind is not impossible in FX markets in light of the existing literature. For example, Evans and Lyons (2005) and Menkhoff et al. (2016) show that order flows have predictive information about FX rates over horizons up to one month. To a lesser extent, Rime et al. (2010) and King et al. (2010) show that order flows have predictive information over horizons up to one week.

The top panel of Figure 2 shows the total open interest in U.S. dollars stacked over the seven largest FX futures: euro, British pound, Swiss franc, Japanese yen, Canadian dollar, Australian dollar, and New Zealand dollar. The euro data include the Deutsche mark data.⁹ The open interest aggregate summed over the selected futures grew about 8.5% per annum over my sample period. The FX futures market grew particularly during expansionary periods such as 2003–07 and 2010–18, and shrank during recessionary periods such as 2000–01 and 2008–09.

The middle panel of Figure 2 breaks down the open interest aggregate by the selected futures. For ease of exposition, the sample period is divided into three subsamples: 1988–2002, 2003–09, and 2010–18.¹⁰ The euro and Australian dollar futures increased their shares over time, whereas the Swiss franc, Japanese yen, and Canadian dollar futures lost their shares. In the third subsample, the three largest futures account for 42% (euro), 18% (Japanese yen), and 14% (British pound) of the market, respectively, and the three commodity-currency futures make up 10% (Canadian dollar), 9% (Australian dollar), and 2% (New Zealand dollar),

⁹ The open interest on French franc futures was so tiny relative to that of the Deutsche mark futures that accounting for the former does not affect the results given in this paper.

¹⁰ This ad-hoc division is motivated by the rise and fall of the futures market growth.

respectively.¹¹

The bottom panel of Figure 2 breaks down the open interest aggregate by different investor types. Speculators increased their shares over time, whereas non-reportable players reduced their shares, a trend suggesting that the threshold levels did not catch up with the open interest growth. On average, speculators, hedgers, and non-reportable players account for 25%, 51%, and 24% of the aggregated market, respectively.

Bid-ask spreads: My analysis accounts for transaction costs, or bid-ask spreads. Bid-ask spreads can vary by time of a day and depend on whether there is a macro news announcement on that day. As such, I obtain median bid-ask spreads during the floor trading hours from 8:20 a.m. to 3:00 p.m. (EST) on the days of payroll announcements. For comparison, the bid-ask spreads data on spot and three-month forwards are populated at the market close of payroll release days. The futures and OTC market data are drawn from Refinitiv Tick History and Eikon, respectively, over the sample period from January 1996 through December 2018.

Figure 3 presents evolutions of bid-ask spreads on the selected futures compared with those on the OTC contracts. The figure reveals that the futures have far smaller bid-ask spreads for all currencies than the OTC contracts. This cost difference may create an incentive for informed traders to engage in the futures market, despite its small size relative to the OTC FX market.

Futures margins: The payroll news timing strategy that follows accounts for the margin requirement faced by participants in the futures market. I collect the margin data on outright futures trading from the Chicago Mercantile Exchange (CME). As the CME margin is set in U.S. dollars per contract, I convert it into the margin rate—the margin divided by the futures price and by the contract size. The margin data start on different dates across contracts, so I fill the missing data backward until January 1996 for all contracts.

¹¹ Commodity currencies are those for which values are highly dependent on the economies' commodity exports.

Figure 4 presents evolutions of margin rates on the selected futures. The figure shows that margin rates broadly increased during the 2008 financial crisis. The FX margin rates move within a range of 1–3% most of the time, although some margin spikes were triggered by non-U.S. events such as the 2016 Brexit for the British pound and the 2015 euro-peg removal for the Swiss franc. Such low margin rates make it easy to leverage information in the futures market.¹²

2.2 Variable definitions

Dirty and clean FX exposures: Let me introduce two measures of FX exposures that play a key role in my analysis—dirty and clean FX exposures. $\text{LONG}_{i,c,t}$ and $\text{SHORT}_{i,c,t}$ denote the long and short open interest in U.S. dollars of the currency future c held by the investor type i , respectively. These variables directly come from the COT data.¹³ The net open interest is simply the difference between the long and short open interests: $\text{NET}_{i,c,t} = \text{LONG}_{i,c,t} - \text{SHORT}_{i,c,t}$. Note that the net open interest has fewer fluctuations in the earlier part of the sample than in the later part of the sample as the FX market grew substantially over time.

To account for heteroskedasticity in the net open interest, I normalize it by the day’s entire FX futures portfolio held by the investor type i as follows:

$$y_{i,c,t} = \frac{\text{NET}_{i,c,t}}{\sum_{c \in \mathcal{C}} (\text{LONG}_{i,c,t} + \text{SHORT}_{i,c,t})}, \quad (1)$$

where $y_{i,c,t}$ denotes a normalized measure of the currency- c exposure held by the investor type i and $\mathcal{C}$ denotes the set of selected futures. The denominator in Equation (1) is the sum of the long and short open interest aggregates held by the investor type i . The normalization also helps reduce the periodicity in open interest data that arises from the quarterly futures.

¹² Black (1975) and Easley et al. (1998), among others, argue that leverage is a vital factor in informed trading.

¹³ I assume zero exposures on New Zealand dollar futures for the period before January 1999 as no data were reported over that period.

I refer to this measure as a dirty FX exposure because it can be sensitive to uninformed traders' ebb and flow on both the long and short sides.

I now introduce an alternative measure of FX exposure as follows:

$$x_{i,c,t} = \frac{\text{NET}_{i,c,t}}{\sum_{c \in \mathcal{C}} |\text{NET}_{i,c,t}|}, \quad (2)$$

where the denominator is the sum of the net open interests in absolute terms over different futures. I refer to this measure as a clean FX exposure, as the denominator washes away some uninformed trading to the extent that it is balanced between long and short sides. For a given investor type, absolute values of the clean exposures add up to unity over different futures—thus, they resemble portfolio weights.

Table 1 presents the summary statistics of the dirty and clean exposures. Panels A, B, and C correspond to speculators, hedgers, and non-reportable players, respectively. Speculators' and hedgers' exposures have larger standard deviations than those of non-reportable players. The dirty exposures (the left side) have smaller standard deviations than the clean exposures (the right side). Across all investor types, the euro and Japanese yen futures have the highest standard deviations as they make up the largest shares of the futures market.

Payroll-news returns: My analysis uses FX futures' intraday returns surrounding payroll announcements, which I call payroll-news returns. Let $r_{c,t}^{(b,a)}$ denote the payroll-news return between b minutes before a payroll release (8:30 a.m.) and a minutes after the release, where c indexes the currency future and t indexes the payroll release date. The intraday returns data come from Refinitiv Tick History over the period from January 1996 through December 2018. The Deutsche mark future is used in lieu of the euro future for the period before January 1999. All of the CME futures represent the value of a non-U.S. currency in U.S. dollars, so a positive (negative) return means an appreciation (depreciation) of the non-U.S. currency relative to the U.S. dollar.

Dirty exposures leading up to large payroll-news returns: To examine the

possibility of informed trading in FX futures, I look into the FX exposures leading up to large positive or negative payroll-news returns. These variables are defined as follows. Let $\mathcal{T}_+^c$ denote the set of payroll release days where payroll-news returns are greater than the full-sample standard deviation with a fixed intraday window. That is, $\mathcal{T}_+^c = \{t \in \mathcal{T} | r_{c,t}^{(b,a)} > std(r_{c,t}^{(b,a)})\}$, where $\mathcal{T}$ denotes the set of all payroll release days between January 1996 and December 2018 and std stands for the standard deviation. I then compute an average of the dirty FX exposures leading up to the large positive payroll-news returns, denoted by $y_{i,c}^+$, as follows:

$$y_{i,c}^+ = \frac{1}{|\mathcal{T}_+^c|} \sum_{t \in \mathcal{T}_+^c} y_{i,c,t-3d}, \quad (3)$$

where $3d$ stands for three calendar days. Note that this variable is computed using the three-day-lagged (Tuesday) exposures data immediately preceding a payroll announcement (Friday). Strictly speaking, the set $\mathcal{T}_+^c$ and the variable $y_{i,c}^+$ depend on the choice of an intraday window but I suppress such dependence for notational convenience.

Similarly, let $\mathcal{T}_-^c$ denote the set of payroll release days where the payroll-news returns are more negative than the full-sample standard deviation: $\mathcal{T}_-^c = \{t \in \mathcal{T} | r_{c,t}^{(b,a)} < -std(r_{c,t}^{(b,a)})\}$. I compute an average of the dirty FX exposures leading up to the large negative payroll-news returns, denoted by $y_{i,c}^-$, as follows:

$$y_{i,c}^- = \frac{1}{|\mathcal{T}_-^c|} \sum_{t \in \mathcal{T}_-^c} y_{i,c,t-3d}. \quad (4)$$

A significant difference between $y_{i,c}^+$ and $y_{i,c}^-$ would suggest the possibility of informed trading about payroll news. For example, the investor type with a negative payroll news—or a positive news for FX futures—is likely to have $y_{i,c}^+$ that is greater than $y_{i,c}^-$.

Dirty and clean portfolio earnings around payroll news: I compute two measures of intraday returns that each investor type might have earned on their entire FX futures portfolio around payroll announcements—dirty and clean portfolio earnings. The first measure

is defined using the dirty exposures as follows:

$$de_i^{(b,a)} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{c \in \mathcal{C}} r_{c,t}^{(b,a)} y_{i,c,t-3d}, \quad (5)$$

where $de_i^{(b,a)}$ denotes the dirty portfolio earnings over an intraday window starting b minutes before payroll releases and ending a minutes after the releases. Note that the portfolio earnings are computed using the three-day-lagged exposures data immediately preceding payroll announcements. This earning measure approximates the portfolio return that each investor type might have earned with respect to payroll news if the position changes between the most recent snapshot date and the payroll release time had been orthogonal to the subsequent payroll surprise.

I now introduce an alternative measure of portfolio earnings using the clean exposures as follows:

$$ce_i^{(b,a)} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{c \in \mathcal{C}} r_{c,t}^{(b,a)} x_{i,c,t-3d}, \quad (6)$$

where $ce_i^{(b,a)}$ denotes the clean portfolio earning by the investor type i .

The dirty and clean portfolio earnings will provide direct evidence as to which investor type represents informed traders or liquidity providers. An investor type with positive (negative) earnings is likely to be informed traders (liquidity traders) with respect to payroll news.

Payroll surprises: The U.S. payroll forecast data come from Action Economics over the sample period from January 1988 to December 2018. Using the consensus forecast data, I define a standardized payroll surprise as

$$\text{SURP}_t = \frac{A_t - F_t - \text{mean}(A_t - F_t)}{\text{std}(A_t - F_t)}, \quad (7)$$

where A_t and F_t refer to the actual release and the consensus forecast, respectively, and *mean* stands for the mean. A positive surprise indicates better than expected U.S. employment so

it is generally accompanied by negative payroll-news returns on FX futures.

FX portfolio margin rate: Part of my analysis calls for an intraday portfolio margin rate, but such data are rarely available. Instead, I take a proxy of a portfolio margin rate by summing overnight margin rates across different contracts:

$$m_t = \sum_{c \in \mathcal{C}} w_{c,t} m_{c,t}, \quad (8)$$

where $m_{c,t}$ denotes the currency-specific margin rate at time t and m_t denotes the margin rate for an FX portfolio with a weight of $w_{c,t}$. Note that this calculation will (considerably) overestimate an intraday portfolio margin rate for two reasons. First, the overnight margins used in this paper are larger than the intraday margins that I am supposed to use. Second, the summation does not account for diversification effects or inter-commodity margin credits. Because of the overestimation, the analysis that follows takes a conservative perspective.

3 Informed trading in FX futures

This section provides evidence of informed trading in the futures market, highlighting the different roles of speculators and hedgers. Specifically, I document that certain types of investors have invested in FX futures in the right direction ahead of payroll announcements and thus might have earned significant gains from their FX portfolios around payroll announcements. I also discuss the long-lived nature of informed trading in FX futures.

3.1 FX exposures ahead of payroll announcements

In standard microstructure theory, information can manifest itself into investors' trades or positions before it is released. To study informed trading in FX futures, I examine whether some investors have taken FX exposures in the right direction ahead of payroll announcements.

Figure 5 compares the average dirty FX exposures leading up to large positive payroll-news returns, $y_{i,c}^+$, to those leading up to large negative payroll-news returns, $y_{i,c}^-$. Here, $y_{i,c}^+$ and $y_{i,c}^-$ are obtained with an intraday window that starts five minutes before payroll announcements and ends 30 minutes after the announcements. The figure shows the average exposure difference, $\Delta = y_{i,c}^+ - y_{i,c}^-$, for each futures contract as well as its statistical difference from zero.¹⁴

The top panel shows that speculators are more likely to buy FX futures before positive payroll-news returns than before negative payroll-news returns. The exposure differences are most pronounced for the euro ($\Delta = 4.71\%$), Japanese yen ($\Delta = 2.62\%$), Australian dollar ($\Delta = 1.54\%$), and Canadian dollar ($\Delta = 1.29\%$) in a decreasing order. The differences are statistically significant at the 5 or 10% level. This empirical regularity suggests that speculators have invested in FX futures in the right direction ahead of payroll announcements rather than in the wrong direction—evidence that speculators appear to have had superior information about the impending payroll news.

The middle panel shows that hedgers have held opposite positions from speculators. Hedgers are more likely to sell FX futures before positive payroll-news returns than before negative payroll-news returns, except for the New Zealand dollar future. The exposure differences are most pronounced for the euro ($\Delta = -6.11\%$), Japanese yen ($\Delta = -4.00\%$), Australian dollar ($\Delta = -2.04\%$), and Canadian dollar ($\Delta = -1.56\%$). The differences are statistically significant at the 5 or 10% level. This pattern suggests that hedgers have invested in FX futures in the wrong direction rather than in the right direction.

The bottom panel shows that non-reportable players are more likely to buy FX futures before positive payroll-news returns than before negative payroll-news returns. The exposure differences are statistically significant for Japanese yen and Australian dollar futures at the 5% level. In light of my finding, the non-reportable category appears to encompass not only small retail and corporate investors but also some skilled investors, who do not meet

¹⁴ I apply a t -test of the null hypothesis that two samples have the same mean.

minimum threshold levels.

Overall, speculators—and non-reportable players to a lesser extent—appear to have had better payroll information. In contrast, hedgers seem to have provided liquidity in the FX futures market. The evidence is stronger for the contracts with relatively large net exposures, such as euro, Japanese yen, and Australian dollar futures. These patterns are qualitatively similar across different choices of the intraday window over which payroll-news returns are calculated (not shown).

3.2 FX portfolio earnings around payroll news

I have shown some evidence of informed trading by looking into investors' pre-payroll-news FX exposures, although the evidence is not always statistically significant at the currency level. In this section, I will provide more conclusive evidence of informed trading by examining investors' portfolio earnings around payroll announcements. Note that the portfolio earnings introduced in this paper are net-exposure-weighted so that it is important to have correct exposures in contracts with large net exposures to identify informed trading at the portfolio level.

Figure 1 presents dirty portfolio earnings around payroll news, as defined in Equation (5). For better visualization, I here assume a wide intraday window that starts 30 minutes before payroll announcements and lasts until three hours after the announcements. Panels A, B, and C correspond to speculators, hedgers, and non-reportable players, respectively. Striking patterns emerge with respect to investors' portfolio earnings. Speculators' earnings jump upward immediately after the 8:30 a.m. payroll announcements and reach 1.5 basis points an hour after the announcements. In contrast, hedgers' earnings plunge immediately after payroll announcements and reach about -2 basis points an hour after the announcements. Non-reportable players' earnings exhibit a pattern similar to those of speculators, although the former are smaller than the latter. The sizes of the jumps at 8:30 a.m., either

gains or losses, are a few times larger in clean portfolio earnings (not shown) than in dirty portfolio earnings (shown).

Table 2 presents the summary statistics and Sharpe ratios of the portfolio earnings with a fixed opening time ($b = 5$) and various closing times ($a \in \{30, 60, 120, 180\}$). Speculators appear to have earned positive and statistically significant returns, with annualized Sharpe ratios of 0.61 to 0.70 (dirty earnings) and 0.55 to 0.64 (clean earnings). In stark contrast, hedgers appear to have incurred statistically significant losses, with Sharpe ratios of -0.54 to -0.64 (dirty earnings) and -0.46 to -0.58 (clean earnings). Non-reportable players' earnings have smaller Sharpe ratios than those of speculators.

Overall, speculators—and non-reportable players to a lesser extent—appear to have earned statistically significant FX portfolio gains during payroll announcements, whereas hedgers appear to have incurred statistically significant losses. As such, speculators have acted like informed traders, whereas hedgers have provided liquidity. In Appendix A, I show that speculators' exposures have led hedgers' exposures using Granger causality tests, rather than vice versa. That is, speculators' exposures are largely driven by their own eagerness to trade, whereas hedgers' exposures are dependent on speculators' lagged exposures. I thus consider the leadership of speculators as additional evidence that they are informed traders.

Two remarks are in order. I first acknowledge that this paper does not identify the source of informed trading. Nevertheless, there are three possible explanations for speculators' earnings associated with payroll announcements. First, the recent literature documents evidence suggesting the possibility of information leakage about macroeconomic announcements. For example, Bernile et al. (2016) find that the pre-FOMC abnormal order imbalances in S&P 500 futures are aligned with post-FOMC policy surprises.¹⁵ Kurov et al. (2019) discover similar pre-announcement drifts before several macroeconomic announcements. Given the existing studies, speculators' earnings may be driven by information leakage. Second, some

¹⁵ Loosely related, Cieslak et al. (2019) find the cyclical pattern of stock returns over the FOMC cycle and attribute the pattern to the informal communication of policy makers with the financial media and markets.

speculators may have access to proprietary payroll data or the ability to collect them on their own. For example, Cajner et al. (2018) show that the payroll data from the largest payroll processing company in the U.S., the ADP, help predict labor market conditions. In a subsequent paper, Cajner et al. (2020) use the ADP data to provide a timely analysis of the job losses in early months of the Covid-19 pandemic. Third, informed trading may be explained by some investors who account for professional forecasters' behavioral bias. For example, while Campbell and Sharpe (2009) report an anchoring bias in consensus forecasts, Zhou (2015) shows that sophisticated traders better predict macroeconomic news by accounting for such behavioral bias.

I also acknowledge that speculators' earnings are uniquely found for payroll announcements. Applying a similar analysis to the FOMC and other key macroeconomic announcements, I find no similar evidence with respect to those announcements (see Appendix B). This uniqueness is consistent with practitioners' beliefs that payrolls are the king of announcements and the existing studies showing that payroll announcements have a bigger effect on Treasury markets than other macroeconomic announcements (see, for example, Gilbert et al. 2017; Huang 2018). Nevertheless, further studies are warranted to understand the specialness of payroll announcements.

3.3 Longevity of informed trading

In the preceding section, I used the same week's (three-day-lagged) FX exposures to look into the behavior of investors' dirty portfolio earnings around payroll announcements. In this section, I repeat a similar analysis using the FX exposures that are observed with various lags. This exercise is interesting because it can reveal the longevity of informed trading in FX futures. For example, if the portfolio earnings based on the FX exposures with a six-week lag were significantly different from zero, it would suggest that informed trading may have started six weeks before the announcements.

Figure 6 presents dirty portfolio earnings that are calculated using the FX exposures with various lags up to 13 weeks. Here, the intraday window is fixed at starting five minutes before announcements and ending 30 minutes after the announcements. Speculators' earnings (Panel A) become significantly positive at the 5% level when the underlying exposure data have a lag of six weeks or fewer. Hedgers' earnings (Panel B) become significantly negative at the 5% level when the underlying exposure data have a lag of eight weeks or fewer. These results suggest that informed trading may have started about eight weeks before announcements.

To further support long-lived information, I examine whether the FX exposure measures contain information about future payroll surprises up to two months ahead. If current FX exposures have predictive power for future payroll surprises, it would suggest that informed traders are likely to have superior information about future payrolls compared with professional forecasters.

Let $PCA_{i,t}^{drt}$ and $PCA_{i,t}^{cln}$ denote the first PCA factor of either dirty or clean exposures across various FX futures held by the investor type i , respectively. For notational convenience, I will henceforth suppress the superscript that indicates the dirty or clean exposures. I then examine the predictive information of the PCA factors for future payroll surprises as follows:

$$SURP_{t+h} = \beta_0 + \beta_1 PCA_{i,t-3d} + \beta_2 TMSP_{t-3d} + \varepsilon_{t+h}, \quad (9)$$

where $TMSP_{t-3d}$ denotes the three-day-lagged term premium defined as a quarterly moving average of the yield spreads between 10-year Treasury bonds and three-month Treasury bills and ε_{t+h} is a forecasting error.¹⁶ The data are populated on the payroll release days (but not the month-end days) and the PCA factor represents the three-day-lagged (Tuesday) common exposure immediately preceding a payroll announcement (Friday). That said, with $h = 0$, the prediction turns to the three-day-ahead forecasting of the impending payroll surprises.

¹⁶ The control variable $TMSP$ is accounted for because it is known to be a good predictor of economic activities (see, for example, Estrella and Hardouvelis 1991; Estrella and Mishkin 1998).

A negative (positive) coefficient on the PCA factor indicates that the investor type is likely informed (liquidity) traders; recall that good U.S. payroll news is accompanied by negative futures returns.

The top panel of Table 3 shows the forecasting power of speculators' PCA factor. The panel shows that the PCA factor is negatively associated with the subsequent payroll surprises, with the statistical significance being stronger with clean exposures (the top-right panel) than with dirty exposures (the top-left panel). This negative relation suggests that speculators have invested in FX futures in the right direction.

The middle panel of Table 3 shows the forecasting power of hedgers' PCA factor. The panel shows that the PCA factor is positively associated with the subsequent payroll surprises, with the statistical significance being stronger with clean exposures than with dirty exposures. This positive relation suggests that hedgers have invested in FX futures in the wrong direction ahead of the subsequent payroll surprises.

The bottom panel of Table 3 shows the forecasting power of non-reportable players' PCA factor. The results are similar to those obtained with speculators' PCA factor. The negative coefficients on the PCA factor suggest that non-reportable players have invested in FX futures in the right direction.

Overall, I find that current FX exposures have information about not only the nearest payroll announcement but also the second nearest one. In other words, part of information contained in current FX exposures will survive even after the next announcement day. This finding is somewhat surprising because some microstructure theory such as that of Kyle (1985) suggests that information should be fully incorporated into asset prices until the end of trading as market makers can learn from informed traders' order flows.

What can explain such long-lasting information? In an effort to understand this issue, I examine measurement errors in the first-release payroll surprises ($SURP_t$). Figure 7 shows the scatter plot between the first- and final-release surprises in my sample period, where

the final-release payroll surprise (SURP_t^f) is defined as the standardized difference between the finally revised payroll number and the consensus forecast. The regression result in the figure shows that only 33% of the variation in SURP_t^f is explained by SURP_t and that a one-unit change in the former is associated with a 0.58-unit change in the latter. Therefore, the first-release payroll data contain a considerable amount of measurement errors, thereby providing very noisy information about current employment conditions.

Given substantial noises in the first-release payroll data, two possibilities are likely to explain long-lived information in FX exposures. First, it is possible that speculators may be patient in executing their trades, probably, waiting for high noise trading activity and market liquidity. This possibility is related to the growing evidence on strategic timings of informed trading. For example, Collin-Dufresne and Fos (2015, 2016) identify informed traders from the Schedule 13D filings and show that the traders account for noise trading activity and market liquidity when executing trades. Authors such as Kacperczyk and Pagnotta (2019) and Ahern (2020) provide evidence of the strategic timings of informed trades. In addition, the futures market allows for limit orders. Strategic timings of trades (and limit orders) can slow down information percolation into FX futures prices. Second, it is also possible that some speculators may have exclusively private information that is difficult for other traders (and professional forecasters) to discern. For example, Foster and Viswanathan (1994) provide a strategic trading model with two informed traders in which one has more information than the other while both share some common information. According to the model, the incorporation of exclusively private information into asset prices can be delayed as the more informed tries to avoid revealing information to the less informed.

3.4 Disaggregated investors' portfolio earnings around payroll news

The investor types considered so far are broad and raise the question of who is truly driving information in the futures market. For example, speculators encompass not only hedge funds

but also any other investors whose trading purpose is not hedging, and hedgers can include large non-financial firms in addition to dealer banks.

In this section, I revisit investors' FX portfolio earnings around payroll announcements using the disaggregated COT data. The disaggregated data shuffle reportable traders—speculators and hedgers—into four sub-categories: asset managers, leveraged funds, dealers, and other reportable players.¹⁷ Asset managers and leveraged funds are buy-side clients of dealers and other reportable players include corporate treasuries, central banks, and non-dealer banks. The disaggregated data only go back to June 2006.

Table 4 presents the summary statistics and Sharpe ratios of disaggregated investors' portfolio earnings. Here, I assume a fixed opening time ($b = 5$) and various closing times ($a \in \{30, 60, 120, 180\}$) to calculate the portfolio earnings. Asset managers (Panel A) appear to have earned positive returns, although the statistical significance is obtained at the 5% or 10% level only with shorter closing times. Leveraged funds (Panel B) also seem to have earned positive returns, although the result is statistically weak or insignificant. One may expect dealers to have earned some gains because they have access to the order flows of their clients and can bring the processed information to the futures market. Contrary to the expectation, dealers (Panel C) appear to have incurred losses that are statistically significant at the 5% (when $a = 60$) or 10% level (when $a = 30$). Finally, other reportable players' earnings (Panel D) turn out to be near zero.

Overall, the disaggregated results suggest that asset managers and leveraged funds appear to have earned some gains during payroll announcements, and dealers appear to have provided liquidity to those traders. Of note, hedge funds are typically captured in the leveraged funds category, taking speculative positions or arbitrage across FX contracts. Given that leveraged funds' portfolio earnings are statistically weak, it is not clear yet whether hedge funds contribute to informed trading in FX futures. In Section 6, I will discuss the role of hedge funds as informed traders.

¹⁷ There is no mapping between the reportable groups and the disaggregated groups.

4 Payroll news timing

I have shown that investors' FX exposures contain unique information about subsequent FX returns around payroll releases. This section introduces a payroll news timing strategy that utilizes investors' FX exposures and evaluates its economic gains against various reference portfolios. I also discuss the dependence of the timing strategy on the U.S. business cycle.

4.1 A framework for payroll news timing

To set the stage, consider a reference portfolio that consists of $(100 - \alpha)\%$ risky assets and $\alpha\%$ cash holdings. This section lays out a payroll news timing strategy that exploits information in investors' FX exposures to boost the performance of the reference portfolio. In essence, the timing strategy executes an FX portfolio trade just before a payroll announcement and unwinds it any time before the day's market close. In addition, cash holding is an inevitable factor in the timing strategy as it is used for the margin required in the intraday FX portfolio trade.

The key idea of the timing strategy lies in constructing a target FX portfolio to engage in around payroll announcements. I propose forming a target FX portfolio by exactly mimicking informed traders' clean exposures that are observed just before a payroll release. For example, one can mimic speculators' or non-reportable players' clean exposures, with a portfolio weight of $w_{c,t} = x_{i,c,t-10d}$. Note that the 10-day-lagged exposures data are used to form a portfolio for release date t because the same week's exposures data are not publicly available until Friday's announcement. Alternatively, one can mimic the converse of hedgers' clean exposure, in which case the currency future c is given a weight of $w_{c,t} = -x_{i,c,t-10d}$.

The ability to implement the timing strategy is subject to the required margin and the amount of cash available for that margin. Here, I use the maximum FX portfolio exposure possible for a given amount of cash and a given portfolio margin rate, which is defined in

Equation (8). That is, the excess return of the timing strategy, denoted by $r_{timed,t}^{(b,a)}$, may be expressed as a weighted sum of the mimicked FX portfolio return and the risky-asset excess return as follows:

$$r_{timed,t}^{(b,a)} = \underbrace{\frac{0.01\alpha}{m_t} \sum_{c \in \mathcal{C}} w_{c,t} r_{c,t}^{(b,a)}}_{\text{Return from mimicked FX portfolio}} + \underbrace{(1 - 0.01\alpha) r_{e,t}}_{\text{Excess return from risky asset}} \quad (10)$$

where $r_{e,t}$ denotes the excess return of the risky reference asset and $1/m_t$ is identical to the maximum leverage. The performance of the timing strategy depends on whose exposures are mimicked and when the mimicked FX portfolio is entered and exited.

When implementing the timing strategy, I consider three types of risky reference assets: the S&P 500 total return index, the Bank of America U.S. corporate bond total return index, and the JP Morgan aggregate commodity total return index.¹⁸ The top panel of Table 5 presents the summary statistics for the monthly excess returns of those indexes, where the risk-free rate is proxied by three-month Treasury rates. As seen in the table, the excess returns have different characteristics across risky assets—a setting to test the robustness of the timing strategy.

Let me emphasize some features of the timing strategy. The FX portfolio formation is model-free and requires only a single snapshot of the FX exposures data immediately preceding a target payroll release. As a result, the timing strategy is out-of-sample by nature and can be implemented in real time.

4.2 Main results

Table 6 presents the Sharpe ratios and economic gains of the payroll timing strategy with various levels of cash holdings between 5% and 20%.¹⁹ The analysis provided in this section

¹⁸ The commodity index data were provided as a courtesy by the JP Morgan.

¹⁹ Using the Form PF data, Kruttli et al. (2017) find that hedge funds have a wide range of cash holdings, with a mean of 16.4%, a median of 7.5%, and a standard deviation of 21.7%. That said, the cash levels assumed in this paper are not unrealistic.

assumes entering a mimicked FX portfolio five minutes before a payroll release and unwinding it one hour after that release. The economic gain is measured by a Sharpe ratio difference between the timing strategy and no timing strategy.

The top panel of Table 6 presents the results of mimicking speculators' clean exposures. The timing strategy yields Sharpe ratio gains of 0.12 to 0.39 for the stock portfolio, 0.28 to 0.35 for the bond portfolio, and 0.12 to 0.44 for the commodity portfolio. The strategy, in general, works better with a greater amount of cash holdings. With low levels of cash holdings, the strategy is most effective for a bond portfolio, which has small standard deviations compared with the other asset classes considered.

The middle panel of Table 6 presents the results of mimicking the converse of hedgers' clean exposures. The timing strategy yields Sharpe ratio gains of 0.12 to 0.34 for the stock portfolio, 0.21 to 0.28 for the bond portfolio, and 0.11 to 0.41 for the commodity portfolio. These gains are slightly smaller than those observed in the top panel, mirroring the fact that hedgers' clean returns have smaller Sharpe ratios than those of speculators (see Table 2).

The bottom panel of Table 6 presents the results of mimicking non-reportable players' clean exposures. Mimicking non-reportable players yields smaller gains than mimicking speculators or hedgers. This result is associated with the fact that non-reportable players' portfolio earnings have the smallest Sharpe ratio in absolute terms (see Table 2).

In summary, I demonstrate that mimicking informed traders or the converse of liquidity providers can improve upon any form of buy-and-hold portfolios. The robust improvement is in part due to the fact that payroll-news returns have little correlations with monthly risky-asset excess returns. Because of diversification effects, the improvement can be achieved even when risky reference assets have greater Sharpe ratios than a mimicked FX portfolio.

4.3 Additional results

I re-examine the economic gains of the payroll timing strategy after accounting for some real-world features of trading. Specifically, I assess the extent to which transaction cost and prolonged unwinding impair the performance of the timing strategy. The results here are limited to mimicking speculators' clean exposures.

Transaction cost: The top panel of Table 7 presents the Sharpe ratios and economic gains of the payroll timing strategy after accounting for round-trip transaction cost, which is proxied by the median bid-ask spreads over the same intraday window as used for the payroll timing. The cost-adjusted Sharpe ratio gains range from 0.09 to 0.26 for the stock portfolio, 0.11 to 0.20 for the bond portfolio, and 0.09 to 0.31 for the commodity portfolio. These gains are about two-thirds of those obtained with no transaction cost (compare Table 7 to Table 6).

Although transaction cost appears to be non-trivial, some improvements may be possible to help minimize transaction cost. For example, one can consider intermittent timing that engages in FX trading only when information in FX exposures is strong enough to compensate for transaction cost. This possibility requires an econometric model to determine whether to turn on the timing strategy, and I will leave it to future research.

Extended unwinding: The timing strategy evaluated so far assumes that the FX exposures entered earlier are unwound all at once. However, it is unlikely that a sufficient amount of contracts is immediately executable at the best price. A more realistic strategy is to smooth out the unwinding to avoid any price pressure effects (and hide private information).

The bottom panel of Table 7 presents the Sharpe ratios and economic gains of the payroll timing strategy with a two-hour exit time. With the new exit time, Sharpe ratio gains range from 0.12 to 0.34 for the stock portfolio, 0.22 to 0.30 for the bond portfolio,

and 0.12 to 0.41 for the commodity portfolio. According to this result, extended unwinding appears to have a very small effect on the performance of the timing strategy.

4.4 Business-cycle dependence

Let me show an interesting feature of the timing strategy. Figure 8 displays the cumulative excess returns of the timing strategy compared to those with no timing strategy, with 10% cash holdings assumed. For illustration, the timing strategy here assumes entering the FX portfolio that mimics speculators' clean exposures five minutes before a payroll release and unwinding it one hour after the release. The shaded areas refer to the NBER-designated recessions included in the sample. The figure shows that most of the improvement was realized during expansionary periods (especially for several years leading up to the onset of recessions) and little was realized during recessionary periods.

To understand the source of the business-cycle dependence, I examine how clean portfolio earnings differ along phases of the U.S. business cycle. Let $ce_{i,t}^{(b,a)}$ denote the clean portfolio earning by the investor type i for a release date t : that is, $ce_{i,t}^{(b,a)} = \sum_{c \in \mathcal{C}} r_{c,t}^{(b,a)} x_{i,c,t-3d}$. I then run the following regression for each investor type i :

$$ce_{i,t}^{(b,a)} = \alpha \text{EXP}_t + \beta(1 - \text{EXP}_t) + \varepsilon_t, \quad (11)$$

where EXP_t denotes a dummy variable that equals one if a release date t belongs to an NBER-designated expansion and zero otherwise. The coefficients α and β capture an average of clean portfolio earnings during expansions and recessions, respectively.

The regression results are provided in Table 8. Here, b is fixed at five (minutes) and a is assumed to vary, with $a \in \{30, 60, 120\}$. Speculators (Panel A) have earned 5 to 7 basis points each announcement only during expansions and the result is statistically significant at the 1% level. Hedgers (Panel B) have lost 5 to 6 basis points each announcement only during expansions and the result is statistically significant at the 1% level. Non-reportable

players (Panel C) have earned 4 to 6 basis points each announcement only during expansions and the result is statistically significant at the 1 to 5% level. Taken together, I find evidence of informed trading only for expansionary periods.

Note that no evidence of informed trading is found for recessionary periods. This result is somewhat surprising because some existing research suggests that macroeconomic information is more actively processed during recessions than during expansions. For example, Kacperczyk et al. (2014) show that professional managers are good at market timing particularly in bad times. Coibion and Gorodnichenko (2015) provide evidence that forecasters update macroeconomic information more frequently in bad times than in good times. Peng et al. (2007) provide evidence suggesting that investors shift limited attention to market-wide information during uncertain times.

How can the lack of evidence during recessions be reconciled with the existing literature? A possible explanation may be associated with safe-haven demands for the U.S. dollar during recessions. The bad news of U.S. economic activity sometimes causes a surprising appreciation of the U.S. dollar and, as a result, can incur a loss to informed traders. It is also possible that the relevance of macroeconomic information changes over time, as implied by the scapegoat model of Bacchetta and Van Wincoop (2004). Related to the latter possibility, I find modest evidence that speculators might have earned some gains around real gross domestic product (GDP) announcements during recessions, although the statistical significance depends on the intraday window chosen (not shown in this paper). Overall, it would be interesting to study the economic driver of the business-cycle dependence.

5 U.S. dollar return forecasting

FX return forecasting has become a long tradition in international finance since the pioneering work of Meese and Rogoff (1983).²⁰ In this section, I document that investors' FX

²⁰ See Rossi (2013) for a survey.

exposures have predictive power for U.S. dollar fluctuations.

5.1 In-sample evidence

To study the information content of FX exposures for U.S. dollar fluctuations, I consider a methodology for aggregating information on FX exposures across different futures. A natural choice would be the first PCA factor of FX exposures. However, the PCA factor is estimated in a way that maximally represents the total variation of input data and thus can contain a common component that has little to do with a target forecast (U.S. dollar returns in this section). Instead of the PCA, I implement a Kelly and Pruitt (2015) three-pass filter such that, for a given investor type, the three-day-lagged exposures data are best aligned with the impending payroll surprise.²¹ With that choice of a factor proxy, the estimated common factor is best positioned to capture payroll information contained in FX exposures.²²

Figure 9 shows the time-series of the three-pass (TP) factors of dirty exposures. The figure shows that the TP factors have strong co-movements across different investor types, suggesting that FX exposure variations are likely to be driven by a single common factor across all investor types. There was a notable moderation in the fluctuations of non-reportable players' factor over the past decade. Keep in mind that the factors exhibit cyclical patterns regardless of investor types.

Let $r_{d,t}$ denote a monthly log return on the trade-weighted U.S. dollar index against major currencies, published by the Federal Reserve. By the index definition, a positive return is associated with an appreciation of the U.S. dollar relative to a basket of major currencies. The summary statistics of the U.S. dollar returns are provided in the middle panel of Table 5.

I examine the predictive power of TP factors for U.S. dollar returns by running the

²¹ The three-pass filter has been increasingly used in financial applications over the PCA, such as Kelly and Pruitt (2013) and Huang et al. (2015).

²² Strictly speaking, the factor is obtained after the second stage of the three-pass filter.

following regression:

$$\frac{1}{h} \sum_{k=1}^h r_{d,t+k} = \alpha + \beta \text{TP}_{i,t} + \gamma' Z_t + \varepsilon_{t+h}, \quad (12)$$

where $\text{TP}_{i,t}$ denotes the TP factor of the investor type i and Z_t denotes a vector of interest rate differences between the United States and other selected countries. The sample period here spans from January 1988 to December 2018. Strictly speaking, the forecasting in Equation (12) is subject to a look-ahead bias because $\text{TP}_{i,t}$ is estimated using the entire sample period.

The top panel of Table 9 shows the in-sample forecasting power of speculators' TP factor for U.S. dollar returns up to three months ahead. A one-standard-deviation shock in the TP factor is followed by a 0.31% appreciation of the U.S. dollar in the next month. The result is statistically significant at the 1% level. The forecasting relationship persists until the subsequent quarter at the 5% level. In contrast, the selected interest rate differences have no forecasting power for U.S. dollar returns.

The middle and bottom panels of Table 9 show the in-sample forecasting power of hedgers' and non-reportable players' TP factors, respectively. Hedgers' TP factor shows nearly the same forecasting power as that of speculators. Non-reportable players' TP factor shows weaker predictive power than those of reportable players.

5.2 Out-of-sample evidence

FX return forecasting is known to be unstable over sample periods and economic regimes. Here, I study the forecasting power of the TP factors in an out-of-sample setting. The entire sample period is divided into two subperiods: the first in-sample estimation period (January 1988–December 1999) and the out-of-sample evaluation period (January 2000–December 2018). The models are recursively estimated at each point in time throughout the out-of-sample evaluation period. To avoid a look-ahead bias, the TP factor is also recursively estimated using the information available at the time of forecasting. The forecasting power,

or out-of-sample R^2 , is measured using a quadratic loss function.

In the top panel of Table 10, I compare the univariate prediction model conditioned on the TP factor to a random walk model with no drift. The statistical significance is evaluated using the McCracken (2007) test, as the two models are nested. The TP factor helps predict the U.S. dollar index returns at the 1% level for a one-month horizon regardless of investor types and at the 1 or 5% level for a three-month horizon. Speculators' and hedgers' TP factors show stronger forecasting power (larger out-of-sample R^2) than that of non-reportable players, particularly for a one-month horizon.

One may wonder how important it is to align FX exposures with payroll surprises in the U.S. dollar return forecasting. To answer this question, I compare the univariate prediction model conditioned on the TP factor to the model conditioned on the PCA factor. The results are shown in the bottom panel of Table 10. The statistical significance is evaluated using the Diebold and Mariano (2002) test, as the two models are non-nested. Speculators' and hedgers' TP factors outperform their PCA counterparts at the 5 to 10% significance level. However, non-reportable players' TP factor does not perform better than the PCA counterpart.

5.3 Summary

Overall, the TP factors are associated with U.S. dollar returns over the next few months in sample and out of sample. Interestingly, the predictive power stands out particularly when the common factor is aligned with payroll surprises through the three-pass filter of Kelly and Pruitt (2015). Accordingly, I interpret my finding as implying that information about future payrolls is a significant source of U.S. dollar fluctuations.

6 Links to hedge funds

To understand the role of hedge funds as informed traders, I study the connection between speculators' FX exposures and hedge funds' stock exposures. The idea goes as follows. There is evidence that hedge funds are good at timing the stock market (see, for example, Chen and Liang 2007). If some hedge funds had superior information about payrolls, it is possible that they might have engaged in both the FX and stock market timings. Related to this possibility, Albuquerque et al. (2008) document that the information extracted from the stock market helps predict FX returns—evidence of cross-market information linkage.

To test for the connection between speculators' FX exposures and hedge funds' stock exposures, I run the following hedge fund return model in which the stock market beta is tied to a TP factor of FX exposures:

$$r_{g,t+1} = \alpha + (\beta + \gamma \text{TP}_{i,t})r_{m,t+1} + \eta' H_{t+1} + \delta r_{g,t} + \varepsilon_{t+1}, \quad (13)$$

where $r_{g,t}$ denotes the excess return of the hedge fund style g , $r_{m,t}$ denotes the stock market excess return, and H_t denotes the vector of the other six Fung and Hsieh (2004) hedge fund factors.²³ Here, I use the Credit Suisse broad hedge fund index as well as its 10 sub-indexes of different styles. The data come from Refinitiv Eikon over the sample period from January 1994 through December 2018. The return model includes a lagged hedge fund return to control for serial correlation in hedge fund excess returns (see the bottom panel of Table 5). The stock timing coefficient γ measures the extent to which stock exposures are linked to FX exposures.

Table 11 shows estimation results of the hedge fund return model using speculators' TP factor. Here, the table shows only the timing coefficient, γ , suppressing other coefficients to save space. The left and right panels correspond to using dirty and clean exposures, respec-

²³ Some data on the hedge fund factors are available at <https://faculty.fuqua.duke.edu/~dah7/HFRFData.htm>.

tively. With the broad index (the first row), a one-standard-deviation shock in speculators' FX exposures is associated with a 0.04 (dirty exposures) or 0.05 (clean exposures) increase in the stock beta. The positivity of γ arises because the common factor is aligned with payroll surprises.

It should be highlighted that the degree of a connection differs by hedge fund styles. Among the 10 sub-indexes, the managed future style exhibits the strongest relationship between speculators' FX exposures and hedge funds' stock exposures, with $\gamma = 0.13$ (dirty exposures) or $\gamma = 0.14$ (clean exposures). Next to the managed future style, the global macro style exhibits the second strongest relationship, with $\gamma = 0.09$ (dirty exposures) or $\gamma = 0.10$ (clean exposures). Some significant relationships are also observed with the convertible arbitrage and multi-strategy styles. The dedicated short bias style shows a negative connection, although the result is not statistically significant.

Overall, speculators' FX exposures are correlated with hedge funds' stock exposures, especially with those classified as managed futures. In an unreported table, a similar connection is found between non-reportable players' FX exposures and hedge funds' stock timing. An explanation for those connections is that hedge funds may drive both FX exposure variations and stock market timing. I consider the evidence to suggest that hedge funds' trading is a contributing factor to information in FX exposures.

7 Conclusion

This paper studies informed trading and liquidity provision in the FX futures market—the market relatively less understood compared with its OTC peers in the literature. I find that speculators such as hedge funds and asset managers are more likely to be sellers than buyers of FX futures ahead of good U.S. payroll news and thus appear to have earned significant gains around payroll announcements. In contrast, hedgers—in particular, dealers—appear to have met demands from speculators, incurring large losses around payroll announcements.

After providing evidence of informed trading, I introduce a payroll news timing strategy that utilizes informed traders' FX exposures and evaluate its economic gain against various reference portfolios. Specifically, the timing strategy attempts to augment the performance of a reference portfolio by executing an intraday FX trade that mimics speculators' FX exposures around payroll announcements. My analysis shows that the strategy can deliver a Sharpe ratio gain of 0.3 to 0.4 over various reference portfolios, including stocks, bonds, and commodities, with some reasonable amount of cash available for the required margin. Note that the strategy is model-free and uses only publicly available information.

Finally, this paper reveals some salient features of informed trading in FX futures, which provide new insights into informed trading and guide some new research directions. In particular, I find that information in current FX exposures persists up to two months ahead. Although I associate the persistence to the possibility of stealth trading or strategic timing, it is still puzzling because the futures positions data are publicly available. This finding also suggests that informed trading may have started far before the release of payroll information. It would be interesting to develop a theoretical model that can explain both information persistence and early informed trading. I also provide evidence that FX informed trading differs along the U.S. business cycle. While this finding is new to the best of my knowledge, further studies are warranted to understand the economic driver of it.

References

- AHERN, K. R. (2020): “Do proxies for informed trading measure informed trading? Evidence from illegal insider trades,” *Review of Asset Pricing Studies*, 10, 397–440.
- AI, H. AND R. BANSAL (2018): “Risk preferences and the macroeconomic announcement premium,” *Econometrica*, 86, 1383–1430.
- ALBUQUERQUE, R., E. DE FRANCISCO, AND L. B. MARQUES (2008): “Marketwide private information in stocks: Forecasting currency returns,” *Journal of Finance*, 63, 2297–2343.
- ANDERSEN, T. G. AND T. BOLLERSLEV (1998): “Deutsche mark-dollar volatility: Intraday activity patterns, macroeconomic announcements, and longer run dependencies,” *Journal of Finance*, 53, 219–265.
- BACCHETTA, P. AND E. VAN WINCOOP (2004): “A scapegoat model of exchange-rate fluctuations,” *American Economic Review*, 94, 114–118.
- (2006): “Can information heterogeneity explain the exchange rate determination puzzle?” *American Economic Review*, 96, 552–576.
- BERGER, D. W., A. P. CHABOUD, S. V. CHERNENKO, E. HOWORKA, AND J. H. WRIGHT (2008): “Order flow and exchange rate dynamics in electronic brokerage system data,” *Journal of International Economics*, 75, 93–109.
- BERNILE, G., J. HU, AND Y. TANG (2016): “Can information be locked up? Informed trading ahead of macro-news announcements,” *Journal of Financial Economics*, 121, 496–520.
- BLACK, F. (1975): “Fact and fantasy in the use of options,” *Financial Analysts Journal*, 31, 36–41.

- CAJNER, T., L. D. CRANE, R. DECKER, A. HAMINS-PUERTOLAS, C. J. KURZ, AND T. RADLER (2018): “Using payroll processor microdata to measure aggregate labor market activity,” Working paper, Federal Reserve Board.
- CAJNER, T., L. D. CRANE, R. A. DECKER, J. GRIGSBY, A. HAMINS-PUERTOLAS, E. HURST, C. KURZ, AND A. YILDIRMAZ (2020): “The US labor market during the beginning of the pandemic recession,” Working paper, National Bureau of Economic Research.
- CAMPBELL, S. D. AND S. A. SHARPE (2009): “Anchoring bias in consensus forecasts and its effect on market prices,” *Journal of Financial and Quantitative Analysis*, 44, 369–390.
- CERRATO, M., N. SARANTIS, AND A. SAUNDERS (2011): “An investigation of customer order flow in the foreign exchange market,” *Journal of Banking & Finance*, 35, 1892–1906.
- CHEN, Y. AND B. LIANG (2007): “Do market timing hedge funds time the market?” *Journal of Financial and Quantitative Analysis*, 42, 827–856.
- CIESLAK, A., A. MORSE, AND A. VISSING-JORGENSEN (2019): “Stock returns over the FOMC cycle,” *Journal of Finance*, 74, 2201–2248.
- COIBION, O. AND Y. GORODNICHENKO (2015): “Information rigidity and the expectations formation process: A simple framework and new facts,” *American Economic Review*, 105, 2644–78.
- COLLIN-DUFRESNE, P. AND V. FOS (2015): “Do prices reveal the presence of informed trading?” *Journal of Finance*, 70, 1555–1582.
- (2016): “Insider trading, stochastic liquidity, and equilibrium prices,” *Econometrica*, 84, 1441–1475.
- DIEBOLD, F. X. AND R. S. MARIANO (2002): “Comparing predictive accuracy,” *Journal of Business and Economic Statistics*, 20, 134–144.

- EASLEY, D., M. O'HARA, AND P. S. SRINIVAS (1998): "Option volume and stock prices: Evidence on where informed traders trade," *Journal of Finance*, 53, 431–465.
- ESTRELLA, A. AND G. A. HARDOUVELIS (1991): "The term structure as a predictor of real economic activity," *Journal of Finance*, 46, 555–576.
- ESTRELLA, A. AND F. S. MISHKIN (1998): "Predicting US recessions: Financial variables as leading indicators," *Review of Economics and Statistics*, 80, 45–61.
- EVANS, M. D. AND R. K. LYONS (2002): "Order flow and exchange rate dynamics," *Journal of Political Economy*, 110, 170–180.
- (2005): "Meese-Rogoff redux: Micro-based exchange-rate forecasting," *American Economic Review*, 95, 405–414.
- (2008): "How is macro news transmitted to exchange rates?" *Journal of Financial Economics*, 88, 26–50.
- FOSTER, F. D. AND S. VISWANATHAN (1994): "Strategic trading with asymmetrically informed traders and long-lived information," *Journal of Financial and Quantitative Analysis*, 29, 499–518.
- FUNG, W. AND D. A. HSIEH (2004): "Hedge fund benchmarks: A risk-based approach," *Financial Analysts Journal*, 60, 65–80.
- GILBERT, T., C. SCOTTI, G. STRASSER, AND C. VEGA (2017): "Is the intrinsic value of a macroeconomic news announcement related to its asset price impact?" *Journal of Monetary Economics*, 92, 78–95.
- HODRICK, R. (1992): "Dividend yields and expected stock returns: Alternative procedures for inference and measurement," *Review of Financial studies*, 5, 357–386.
- HUANG, D., F. JIANG, J. TU, AND G. ZHOU (2015): "Investor sentiment aligned: A powerful predictor of stock returns," *Review of Financial Studies*, 28, 791–837.

- HUANG, X. (2018): “Macroeconomic news announcements, systemic risk, financial market volatility, and jumps,” *Journal of Futures Markets*, 38, 513–534.
- KACPERCZYK, M., S. V. NIEUWERBURGH, AND L. VELDKAMP (2014): “Time-varying fund manager skill,” *Journal of Finance*, 69, 1455–1484.
- KACPERCZYK, M. AND E. S. PAGNOTTA (2019): “Chasing private information,” *Review of Financial Studies*, 32, 4997–5047.
- KANIEL, R. AND H. LIU (2006): “So what orders do informed traders use?” *Journal of Business*, 79, 1867–1913.
- KELLY, B. AND S. PRUITT (2013): “Market expectations in the cross-section of present values,” *Journal of Finance*, 68, 1721–1756.
- (2015): “The three-pass regression filter: A new approach to forecasting using many predictors,” *Journal of Econometrics*, 186, 294–316.
- KING, M., L. SARNO, AND E. SOJLI (2010): “Timing exchange rates using order flow: The case of the Loonie,” *Journal of Banking & Finance*, 34, 2917–2928.
- KLITGAARD, T. AND L. WEIR (2004): “Exchange rate changes and net positions of speculators in the futures market,” *Economic Policy Review*, 10, 17–28.
- KRUTTLI, M. S., P. MONIN, AND S. W. WATUGALA (2017): “Investor concentration, flows, and cash holdings: Evidence from hedge funds,” *Finance and Economics Discussion Series 2017-121*, Board of Governors of the Federal Reserve System.
- KUROV, A., A. SANCETTA, G. STRASSER, AND M. H. WOLFE (2019): “Price drift before US macroeconomic news: Private information about public announcements?” *Journal of Financial and Quantitative Analysis*, 54, 449–479.
- KYLE, A. S. (1985): “Continuous auctions and insider trading,” *Econometrica*, 53, 1315–1335.

- LOVE, R. AND R. PAYNE (2008): “Macroeconomic news, order flows, and exchange rates,” *Journal of Financial and Quantitative Analysis*, 43, 467–488.
- MCCRACKEN, M. W. (2007): “Asymptotics for out of sample tests of Granger causality,” *Journal of Econometrics*, 140, 719–752.
- MEESE, R. A. AND K. ROGOFF (1983): “Empirical exchange rate models of the seventies: Do they fit out of sample?” *Journal of International Economics*, 14, 3–24.
- MENKHOFF, L., L. SARNO, M. SCHMELING, AND A. SCHRIMPF (2016): “Information flows in foreign exchange markets: Dissecting customer currency trades,” *Journal of Finance*, 71, 601–634.
- MUELLER, P., A. TAHBAZ-SALEHI, AND A. VEDOLIN (2017): “Exchange rates and monetary policy uncertainty,” *Journal of Finance*, 72, 1213–1252.
- NEWBY, W. K. AND K. D. WEST (1987): “A simple positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix,” *Econometrica*, 55, 703–708.
- PARK, Y.-H. (2021): “Spread position as a leading economic indicator,” *Forthcoming in Journal of Financial Markets*.
- PENG, L., W. XIONG, AND T. BOLLERSLEV (2007): “Investor attention and time-varying comovements,” *European Financial Management*, 13, 394–422.
- RIME, D., L. SARNO, AND E. SOJLI (2010): “Exchange rate forecasting, order flow and macroeconomic information,” *Journal of International Economics*, 80, 72–88.
- ROSSI, B. (2013): “Exchange rate predictability,” *Journal of Economic Literature*, 51, 1063–1119.
- SAGER, M. AND M. P. TAYLOR (2008): “Commercially available order flow data and exchange rate movements: Caveat emptor,” *Journal of Money, Credit and Banking*, 40, 583–625.

- SAVOR, P. AND M. WILSON (2013): “How much do investors care about macroeconomic risk? Evidence from scheduled economic announcements,” *Journal of Financial and Quantitative Analysis*, 48, 343–375.
- TORNELL, A. AND C. YUAN (2012): “Speculation and hedging in the currency futures markets: Are they informative to the spot exchange rates,” *Journal of Futures Markets*, 32, 122–151.
- WANG, C. (2004): “Futures trading activity and predictable foreign exchange market movements,” *Journal of Banking & Finance*, 28, 1023–1041.
- ZHOU, J. (2015): “Sophisticated trading and market efficiency: Evidence from macroeconomic news announcements,” Working paper, Harvard University.

Table 1: **Summary statistics of dirty and clean FX exposures**

This table shows the summary statistics of the dirty and clean FX exposures—the measures of the currency-specific exposure held by each investor type in the futures market. Panels A, B, and C correspond to speculators, hedgers, and non-reportable players, respectively. The data cover the seven largest FX futures: euro, British pound, Swiss franc, Japanese yen, Canadian dollar, Australian dollar, and New Zealand dollar. The euro data include the Deutsche mark data. The sample spans from January 1988 to December 2018.

	Dirty exposures (%)				Clean exposures (%)			
	Mean	Min	Max	Std.	Mean	Min	Max	Std.
Panel A: Speculators								
EUR	-0.5	-28.8	17.8	8.2	-0.5	-82.5	69.5	31.3
GBP	0.1	-11.9	15.0	3.8	0.1	-44.1	60.2	17.1
CHF	-0.8	-15.1	11.9	3.4	-2.8	-46.0	51.2	15.4
JPY	-2.4	-23.6	18.2	7.2	-8.8	-74.4	75.0	30.0
CAD	0.4	-14.7	13.8	3.6	2.0	-52.1	80.3	17.7
AUD	1.1	-5.8	10.3	2.7	4.5	-26.1	55.5	11.1
NZD	0.3	-2.1	3.0	0.7	1.0	-10.1	17.7	3.2
Panel B: Hedgers								
EUR	-0.4	-23.1	32.3	11.6	-2.0	-73.8	73.5	32.5
GBP	0.2	-15.4	14.4	5.4	1.2	-55.0	40.1	16.9
CHF	1.0	-10.8	12.1	4.5	2.8	-38.4	40.3	13.5
JPY	3.5	-20.4	27.5	9.6	9.0	-66.5	72.1	28.9
CAD	-1.6	-17.3	9.9	4.4	-4.6	-57.1	33.2	14.0
AUD	-1.3	-11.3	10.4	3.3	-3.8	-47.9	29.3	10.1
NZD	-0.2	-2.6	1.7	0.6	-0.6	-11.0	7.6	2.0
Panel C: Non-reportable players								
EUR	1.1	-6.8	12.4	4.2	6.8	-66.6	63.5	29.5
GBP	0.1	-7.6	6.4	2.1	-0.1	-49.1	51.8	15.7
CHF	-0.4	-6.7	6.1	2.5	-4.2	-55.3	51.0	19.0
JPY	-1.1	-10.6	8.9	3.2	-8.7	-75.5	65.0	23.5
CAD	1.2	-5.0	6.0	2.0	9.6	-34.6	61.7	16.3
AUD	0.3	-2.2	3.7	1.0	2.8	-19.2	40.8	9.0
NZD	-0.0	-0.6	0.2	0.1	0.0	-7.7	1.6	0.8

Table 2: **Investors' FX portfolio earnings around payroll news**

This table shows the summary statistics of FX portfolio earnings around payroll news by different investor types. The FX portfolio earnings are the returns that each investor type might have earned on their entire FX futures portfolio over an intraday window that starts b minutes before payroll announcements and ends a minutes after the announcements. For this table, b is fixed at five (minutes) and a is assumed to vary, with $a \in \{30, 60, 120, 180\}$. Panels A, B, and C correspond to speculators, hedgers, and non-reportable players, respectively. The dirty earnings (the left side) and the clean earnings (the right side) are defined in Equations (5) and (6), respectively. The Sharpe ratios are annualized. The sample period covers January 1996 through December 2018. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively.

	Dirty portfolio earnings (bps)			Clean portfolio earnings (bps)		
	Mean	Std.	Sharpe ratio	Mean	Std.	Sharpe ratio
Panel A: Speculators						
$a = 30$	1.13***	6.48	0.61	4.08***	25.61	0.55
$a = 60$	1.48***	7.36	0.70	5.23***	28.35	0.64
$a = 120$	1.58***	8.25	0.66	5.39***	31.88	0.59
$a = 180$	1.61***	8.76	0.64	5.55***	34.03	0.56
Panel B: Hedgers						
$a = 30$	-1.49***	9.55	-0.54	-3.89**	26.87	-0.50
$a = 60$	-1.97***	10.68	-0.64	-5.01***	29.73	-0.58
$a = 120$	-2.04***	11.89	-0.59	-4.99**	33.31	-0.52
$a = 180$	-1.96**	12.56	-0.54	-4.74**	35.30	-0.46
Panel C: Non-reportable players						
$a = 30$	0.46**	3.76	0.42	3.73**	29.06	0.44
$a = 60$	0.59**	4.19	0.49	4.70**	32.34	0.50
$a = 120$	0.67**	4.57	0.51	4.70**	36.34	0.45
$a = 180$	0.62**	4.83	0.44	4.30*	37.72	0.40

Table 3: **Forecasting professional forecasters' errors**

This table shows the information content of FX exposures for future payroll surprises based on the following regression:

$$\text{SURP}_{t+h} = \beta_0 + \beta_1 \text{PCA}_{i,t-3d} + \beta_2 \text{TMSP}_{t-3d} + \varepsilon_{t+h},$$

where SURP_{t+h} denotes the h -month-ahead payroll surprise (the standardized difference between the actual payroll release and the consensus forecast), $\text{PCA}_{i,t}$ denotes the first PCA factor of either dirty (shown in the left side) or clean (shown in the right side) exposures across various FX futures held by the investor type i , TMSP denotes the term premium defined as a quarterly moving average of the yield spreads between ten-year Treasury bonds and three-month Treasury bills, and ε_{t+h} is a forecasting error. Note that the PCA factor represents the three-day-lagged (Tuesday) common FX exposure immediately preceding a payroll announcement (Friday). That said, with $h = 0$, the prediction turns to the three-day-ahead forecasting of the impending payroll surprises. Newey and West (1987) robust t -statistics with an optimal lag are shown in parentheses. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively. The sample spans from January 1988 to December 2018.

Using dirty exposures				Using clean exposures			
	$h = 0$	$h = 1$	$h = 2$		$h = 0$	$h = 1$	$h = 2$
Panel A: Speculators							
PCA	-1.11** (-2.35)	-1.05** (-2.17)	-0.94** (-2.12)	PCA	-0.33*** (-2.83)	-0.29** (-2.15)	-0.25** (-2.11)
TMSP	-0.02 (-0.39)	-0.02 (-0.51)	-0.01 (-0.29)	TMSP	-0.01 (-0.21)	-0.01 (-0.35)	-0.01 (-0.14)
Const.	0.03 (0.26)	0.04 (0.38)	0.01 (0.11)	Const.	0.01 (0.12)	0.03 (0.25)	-0.00 (-0.01)
<i>adj. R</i> ²	0.00	0.00	0.00	<i>adj. R</i> ²	0.01	0.00	0.00
Panel B: Hedgers							
PCA	0.95*** (2.82)	0.99*** (2.88)	0.98*** (2.97)	PCA	0.38*** (3.15)	0.33*** (2.69)	0.32*** (2.84)
TMSP	-0.01 (-0.31)	-0.02 (-0.44)	-0.01 (-0.23)	TMSP	-0.02 (-0.34)	-0.02 (-0.47)	-0.01 (-0.26)
Const.	0.02 (0.20)	0.03 (0.33)	0.01 (0.05)	Const.	0.02 (0.23)	0.03 (0.35)	0.01 (0.08)
<i>adj. R</i> ²	0.01	0.01	0.01	<i>adj. R</i> ²	0.01	0.01	0.01
Panel C: Non-reportable players							
PCA	-2.62*** (-2.79)	-3.12*** (-3.39)	-3.23*** (-3.62)	PCA	-0.40*** (-3.12)	-0.41*** (-3.44)	-0.37*** (-3.28)
TMSP	-0.01 (-0.23)	-0.01 (-0.35)	-0.01 (-0.13)	TMSP	-0.02 (-0.48)	-0.03 (-0.63)	-0.02 (-0.40)
Const.	0.01 (0.13)	0.02 (0.25)	-0.00 (-0.03)	Const.	0.03 (0.34)	0.05 (0.48)	0.02 (0.18)
<i>adj. R</i> ²	0.01	0.02	0.02	<i>adj. R</i> ²	0.01	0.01	0.01

Table 4: Disaggregated investors' portfolio earnings around payroll news

This table shows the summary statistics of FX portfolio earnings around payroll news by disaggregated investor types. The FX portfolio earnings are the returns that each investor type might have earned on their entire FX futures portfolio over an intraday window that starts b minutes before payroll announcements and ends a minutes after the announcements. For this table, b is fixed at five (minutes) and a is assumed to vary, with $a \in \{30, 60, 120, 180\}$. The dirty earnings (the left side) and the clean earnings (the right side) are defined in Equations (5) and (6), respectively. Panels A, B, C, and D correspond to asset managers, leveraged funds, dealers, and other reportable players, respectively. The Sharpe ratios are annualized. The sample period spans from June 2006 through December 2018. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively.

	Dirty portfolio earnings (bps)			Clean portfolio earnings (bps)		
	Mean	Std.	Sharpe ratio	Mean	Std.	Sharpe ratio
Panel A: Asset managers						
$a = 30$	0.40*	3.63	0.38	3.60*	30.72	0.41
$a = 60$	0.50*	3.92	0.44	4.82**	32.45	0.51
$a = 120$	0.25	4.14	0.21	2.70	35.57	0.26
$a = 180$	0.23	4.09	0.19	2.13	36.10	0.20
Panel B: Leveraged funds						
$a = 30$	0.67	6.01	0.39	2.23	25.56	0.30
$a = 60$	1.00*	6.74	0.52	3.04	28.11	0.37
$a = 120$	0.79	7.80	0.35	1.85	33.33	0.19
$a = 180$	0.57	8.11	0.24	1.32	34.42	0.13
Panel C: Dealers						
$a = 30$	-1.59*	10.52	-0.52	-4.30*	28.47	-0.52
$a = 60$	-2.16**	11.68	-0.64	-5.37**	31.19	-0.60
$a = 120$	-1.57	13.73	-0.40	-3.64	35.80	-0.35
$a = 180$	-1.25	14.04	-0.31	-3.08	37.29	-0.29
Panel D: Other reportable players						
$a = 30$	0.02	2.61	0.03	0.87	26.95	0.11
$a = 60$	0.00	3.09	0.00	0.61	30.12	0.07
$a = 120$	0.19	3.43	0.20	2.67	31.91	0.29
$a = 180$	0.12	3.22	0.13	2.57	31.14	0.29

Table 5: **Summary statistics of (excess) asset returns**

Panel A shows the summary statistics of the excess returns of three risky reference assets: the S&P 500 total return index, the Bank of America U.S. corporate bond total return index, and the JP Morgan aggregate commodity total return index, where the risk-free rate is proxied by three-month Treasury rates. Panel B shows the summary statistics of monthly log returns on the trade-weighted U.S. dollar index against major currencies, published by the Federal Reserve Board. Panel C shows the summary statistics of the excess returns of the Credit Suisse broad hedge fund index as well as its 10 sub-indexes of different styles. All the data have monthly frequency.

	Mean (%)	Min	Max	Std.	Skew.	Kurt.	AR(1)
Panel A: Excess returns of risky reference assets							
S&P 500	0.51	-23.67	21.07	4.84	-0.66	6.48	-0.05
BoA Corp. Bond	0.27	-9.02	5.94	1.59	-0.75	7.65	0.03
JPM Commodity	0.11	-25.42	13.55	5.26	-0.83	5.59	0.13
Panel B: Returns of the U.S. dollar index							
Dollar index	0.01	-5.44	5.25	2.00	0.12	2.83	0.12
Panel C: Excess returns of hedge fund indexes							
Total hedge fund	0.41	-7.96	8.10	1.93	-0.24	6.29	0.19
Convertible arbitrage	0.32	-12.66	5.80	1.77	-2.60	21.37	0.53
Dedicated short bias	-0.95	-100.04	22.30	7.57	-7.90	107.14	0.05
Emerging markets	0.39	-23.44	16.06	3.79	-0.85	9.80	0.30
Market neutral	0.17	-40.49	3.65	2.62	-12.61	196.20	0.05
Event driven	0.43	-12.18	4.21	1.71	-2.02	13.19	0.35
Fixed income arbitrage	0.22	-14.11	4.32	1.45	-4.65	39.86	0.52
Global macro	0.58	-11.90	10.15	2.47	0.03	8.14	0.08
Long short	0.49	-11.84	12.58	2.56	-0.08	7.04	0.18
Managed future	0.20	-9.79	9.54	3.28	0.03	2.92	0.00
Multi-strategy	0.40	-7.49	4.27	1.40	-1.65	9.56	0.29

Table 6: **Economic gains of payroll news timing: Main results**

This table presents the Sharpe ratios and economic gains of the payroll timing strategy with various levels of cash holdings (α) between 5% and 20%. The table also shows the Sharpe ratios of reference portfolios, with $\alpha = 0\%$. Panels A, B, and C correspond to mimicking speculators', hedgers', and non-reportable players' clean exposures, respectively. The timing strategy assumes entering the mimicked FX portfolio five minutes before a payroll release and unwinding it one hour after that release. Three types of risky reference assets are considered: the S&P 500 total return index, the Bank of America U.S. corporate bond total return index, and the JP Morgan aggregate commodity total return index. The economic gain is measured by a Sharpe ratio difference between the timing strategy and no timing strategy. The sample period covers January 1996 through December 2018.

Cash holdings (α)	Sharpe ratios					Sharpe ratio differences			
	0%	5%	10%	15%	20%	5%	10%	15%	20%
	(1)	(2)	(3)	(4)	(5)	(2)-(1)	(3)-(1)	(4)-(1)	(5)-(1)
Panel A: Mimicking speculators									
S&P 500	0.36	0.49	0.60	0.69	0.75	0.12	0.24	0.33	0.39
BoA Corp. Bond	0.59	0.88	0.94	0.92	0.89	0.28	0.35	0.33	0.30
JPM Commodity	0.07	0.19	0.31	0.42	0.51	0.12	0.24	0.35	0.44
Panel B: Mimicking hedgers									
S&P 500	0.36	0.48	0.58	0.66	0.71	0.12	0.22	0.30	0.34
BoA Corp. Bond	0.59	0.84	0.87	0.84	0.80	0.24	0.28	0.25	0.21
JPM Commodity	0.07	0.18	0.30	0.40	0.48	0.11	0.23	0.33	0.41
Panel C: Mimicking non-reportable players									
S&P 500	0.36	0.46	0.54	0.59	0.62	0.10	0.18	0.23	0.26
BoA Corp. Bond	0.59	0.77	0.76	0.71	0.67	0.18	0.17	0.12	0.08
JPM Commodity	0.07	0.17	0.26	0.34	0.40	0.10	0.19	0.27	0.33

Table 7: **Economic gains of payroll news timing: Additional results**

Panel A presents the Sharpe ratios and economic gains of mimicking speculators' clean exposures after accounting for round-trip transaction costs, and Panel B presents the Sharpe ratios and economic gains of the mimicking after accounting for extended unwinding. Cash holdings (α) are assumed to vary between 5% and 20%. The table also shows the Sharpe ratios of reference portfolios, with $\alpha = 0\%$. Three types of risky reference assets are considered: the S&P 500 total return index, the Bank of America U.S. corporate bond total return index, and the JP Morgan aggregate commodity total return index. The economic gain is measured by a Sharpe ratio difference between the timing strategy and no timing strategy. The sample period covers January 1996 through December 2018. Panel A assumes entering the mimicked FX portfolio five minutes before a payroll release and unwinding it one hour after that release. Panel B assumes entering the mimicked FX portfolio five minutes before a payroll release and unwinding it two hours after that release.

Cash holdings (α)	Sharpe ratios					Sharpe ratio differences			
	0%	5%	10%	15%	20%	5%	10%	15%	20%
	(1)	(2)	(3)	(4)	(5)	(2)-(1)	(3)-(1)	(4)-(1)	(5)-(1)
Panel A: Accounting for transaction cost									
S&P 500	0.36	0.45	0.53	0.59	0.62	0.09	0.17	0.22	0.26
BoA Corp. Bond	0.59	0.78	0.79	0.75	0.70	0.19	0.20	0.16	0.11
JPM Commodity	0.07	0.15	0.24	0.32	0.38	0.09	0.17	0.25	0.31
Panel B: Accounting for extended unwinding									
S&P 500	0.36	0.48	0.58	0.66	0.70	0.12	0.22	0.29	0.34
BoA Corp. Bond	0.59	0.86	0.89	0.85	0.81	0.27	0.30	0.26	0.22
JPM Commodity	0.07	0.19	0.30	0.40	0.48	0.12	0.23	0.34	0.41

Table 8: **The business-cycle dependence of clean portfolio earnings**

This table shows the regression results of clean portfolio earnings onto NBER-designated expansion and recession indicators as follows:

$$ce_{i,t}^{(b,a)} = \alpha \text{EXP}_t + \beta(1 - \text{EXP}_t) + \varepsilon_t,$$

where $ce_{i,t}^{(b,a)}$ denotes the clean portfolio earning by the investor type i for a release date t and EXP_t denotes a dummy variable that equals one if a release date t belongs to an NBER-designated expansion and zero otherwise. b is fixed at five (minutes) and a is assumed to vary, with $a \in \{30, 60, 120\}$. The coefficients α and β capture an average of clean portfolio earnings during expansions and recessions, respectively. Newey and West (1987) robust t -statistics with an optimal lag are shown in parentheses. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively. The sample period covers January 1996 through December 2018.

	EXP_t	$1 - \text{EXP}_t$	$adj. R^2$
Panel A: Speculators			
$a = 30$	0.05*** (3.32)	-0.03 (-0.69)	0.00
$a = 60$	0.06*** (3.73)	-0.04 (-1.06)	0.01
$a = 120$	0.07*** (3.49)	-0.08 (-1.51)	0.01
Panel B: Hedgers			
$a = 30$	-0.05*** (-2.99)	0.03 (0.52)	0.00
$a = 60$	-0.06*** (-3.53)	0.05 (0.94)	0.01
$a = 120$	-0.06*** (-3.13)	0.08 (1.45)	0.01
Panel C: Non-reportable players			
$a = 30$	0.04** (2.46)	0.00 (0.05)	-0.00
$a = 60$	0.06*** (2.95)	-0.03 (-0.70)	0.00
$a = 120$	0.06** (2.56)	-0.06 (-0.98)	0.00

Table 9: U.S. dollar return forecasting: In-sample

This table shows the predictive power of three-pass (TP) factors for U.S. dollar returns by running the following regression:

$$\frac{1}{h} \sum_{k=1}^h r_{d,t+k} = \alpha + \beta \text{TP}_{i,t} + \gamma_1(i_t^{DE} - i_t^{US}) + \gamma_2(i_t^{JP} - i_t^{US}) + \gamma_3(i_t^{CA} - i_t^{US}) + \gamma_4(i_t^{AU} - i_t^{US}) + \varepsilon_{t+h},$$

where $r_{d,t}$ denotes a monthly log return on the trade-weighted U.S. dollar index against major currencies, $\text{TP}_{i,t}$ denotes the three-pass factor of FX exposures of the investor type i , and the control variables include the short-term interest rate differences between the United States and four selected countries: Germany (DE), Japan (JP), Canada (CA), and Australia (AU). Panels A, B, and C correspond to speculators, hedgers, and non-reportable players, respectively. Robust t statistics are computed using the Newey and West (1987) method with an optimal lag for one-month horizons and using the Hodrick (1992) method for horizons longer than one month to control for overlapping data. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively. The sample spans from January 1988 to December 2018.

Horizon	$\text{TP}_{i,t}$	$i_t^{DE} - i_t^{US}$	$i_t^{JP} - i_t^{US}$	$i_t^{CA} - i_t^{US}$	$i_t^{AU} - i_t^{US}$	constant	adj. R^2
Panel A: Speculators							
$h = 1$	0.31*** (2.93)	0.07 (0.91)	-0.08 (-1.11)	-0.12 (-1.02)	0.05 (0.64)	-0.18 (-0.74)	0.02
$h = 2$	0.24** (2.46)	0.06 (0.73)	-0.07 (-0.92)	-0.11 (-0.88)	0.04 (0.52)	-0.14 (-0.59)	0.02
$h = 3$	0.19** (2.11)	0.04 (0.50)	-0.05 (-0.77)	-0.08 (-0.65)	0.02 (0.29)	-0.10 (-0.41)	0.02
Panel B: Hedgers							
$h = 1$	0.31*** (3.05)	0.07 (0.82)	-0.07 (-1.06)	-0.13 (-1.04)	0.06 (0.77)	-0.20 (-0.81)	0.02
$h = 2$	0.23** (2.45)	0.05 (0.66)	-0.06 (-0.87)	-0.11 (-0.90)	0.05 (0.61)	-0.15 (-0.63)	0.02
$h = 3$	0.18** (2.04)	0.04 (0.43)	-0.05 (-0.72)	-0.08 (-0.67)	0.03 (0.37)	-0.11 (-0.44)	0.02
Panel C: Non-reportable players							
$h = 1$	0.24** (2.49)	0.06 (0.69)	-0.05 (-0.77)	-0.12 (-1.02)	0.05 (0.69)	-0.14 (-0.58)	0.01
$h = 2$	0.18** (2.04)	0.05 (0.58)	-0.05 (-0.67)	-0.11 (-0.88)	0.04 (0.56)	-0.12 (-0.48)	0.01
$h = 3$	0.16* (1.87)	0.03 (0.39)	-0.04 (-0.60)	-0.08 (-0.64)	0.03 (0.33)	-0.09 (-0.35)	0.01

Table 10: **U.S. dollar return forecasting: Out-of-sample**

This table shows the out-of-sample forecasting power of the three-pass (TP) factors of FX exposures for U.S. dollar index returns. In Panel A, I compare the univariate prediction model conditioned on the TP factor to a random walk model with no drift. The statistical significance is evaluated using the McCracken (2007) test as the two models are nested. In Panel B, I compare the univariate prediction model conditioned on the TP factor to the model conditioned on the PCA factor. The statistical significance is evaluated using the Diebold and Mariano (2002) test as the two models are non-nested. The entire sample period is divided into two subperiods: the first in-sample estimation period (January 1988–December 1999) and the out-of-sample evaluation period (January 2000–December 2018). The models are recursively estimated at each point in time throughout the out-of-sample evaluation period. To avoid look-ahead bias, the TP and the PCA factors are also recursively estimated using the information available at the time of forecasting. The out-of-sample R^2 is measured using a quadratic loss function. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively.

	1 month ahead		3 months ahead	
Panel A: TP factor versus random walk				
	$R^2(\%)$	McCracken	$R^2(\%)$	McCracken
Speculators	2.0	1.43***	1.7	0.89**
Hedgers	2.3	1.57***	1.7	0.89**
Non-reportable players	1.3	1.60***	1.7	1.32***
Panel B: TP factor versus PCA factor				
	$R^2(\%)$	Diebold-Mariano	$R^2(\%)$	Diebold-Mariano
Speculators	0.9	1.56*	1.7	1.94**
Hedgers	0.8	1.73**	1.0	1.54*
Non-reportable players	-0.3	-0.74	0.6	0.52

Table 11: **Connection to hedge funds' stock timing**

This table shows the estimation results of the following hedge fund return model in which the stock market beta is tied to a three-pass (TP) factor of FX exposures:

$$r_{g,t+1} = \alpha + (\beta + \gamma \text{TP}_{i,t})r_{m,t+1} + \eta' H_{t+1} + \delta r_{g,t} + \varepsilon_{t+1},$$

where $\text{TP}_{i,t}$ denotes the three-pass factor of FX exposures of the investor type i , $r_{g,t}$ denotes the excess return of the hedge fund style g , $r_{m,t}$ denotes the stock market excess return, and H_t denotes the vector of the other six Fung and Hsieh (2004) hedge fund factors. Here, I use the Credit Suisse broad hedge fund index as well as its 10 sub-indexes of different styles. The table shows only the timing coefficient, γ , suppressing other coefficients to save space. The result shown here is based on speculators' FX exposures. Newey and West (1987) robust t -statistics with an optimal lag are shown in parentheses. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively. The sample period covers January 1994 through December 2018.

	TP factor of dirty exposures		TP factor of clean exposures	
	γ	t statistic	γ	t statistic
Total hedge fund	0.04*	(1.91)	0.05**	(2.43)
Convertible arbitrage	0.05**	(2.31)	0.06***	(2.60)
Dedicated short bias	-0.04	(-0.61)	-0.08	(-0.82)
Emerging markets	-0.00	(-0.00)	0.02	(0.52)
Market neutral	0.03	(1.19)	0.04	(1.42)
Event driven	0.03	(1.07)	0.04*	(1.73)
Fixed income arbitrage	0.02	(1.35)	0.02	(1.52)
Global macro	0.09**	(2.26)	0.10***	(2.82)
Long short	0.03	(1.08)	0.04	(1.31)
Managed future	0.13**	(2.39)	0.14***	(3.31)
Multi-strategy	0.04*	(1.80)	0.05**	(2.28)

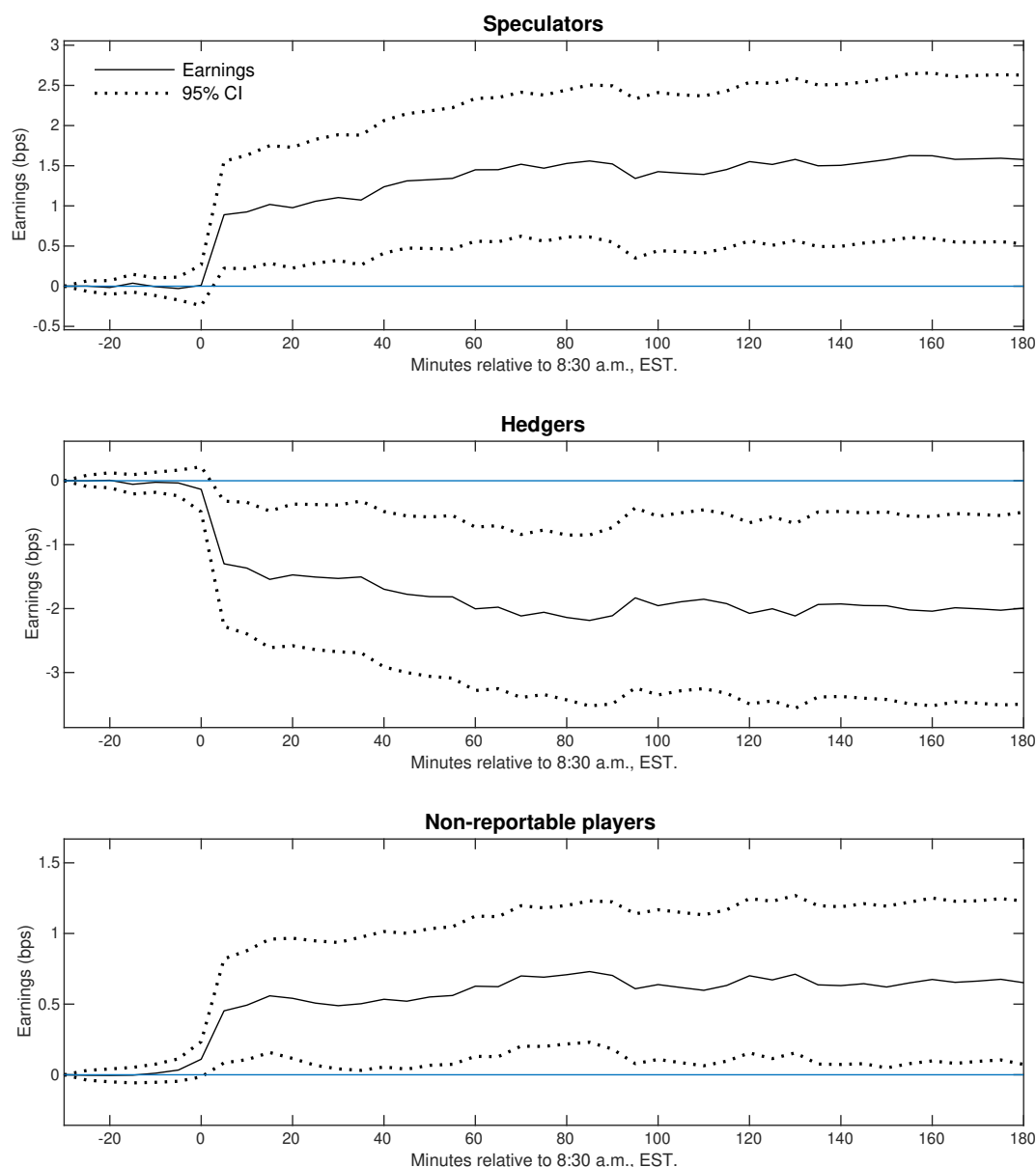


Figure 1: **Dirty portfolio earnings around payroll news by different investor types.** The dirty portfolio earnings approximate the returns that each investor type might have earned on their entire FX portfolio around payroll announcements under some assumptions. The intraday window for computing the portfolio earnings starts 30 minutes before payroll announcements and lasts until three hours after the announcements. The dotted line represents the 95% confidence interval. The portfolio earnings are averaged over the sample period from January 1996 through December 2018.

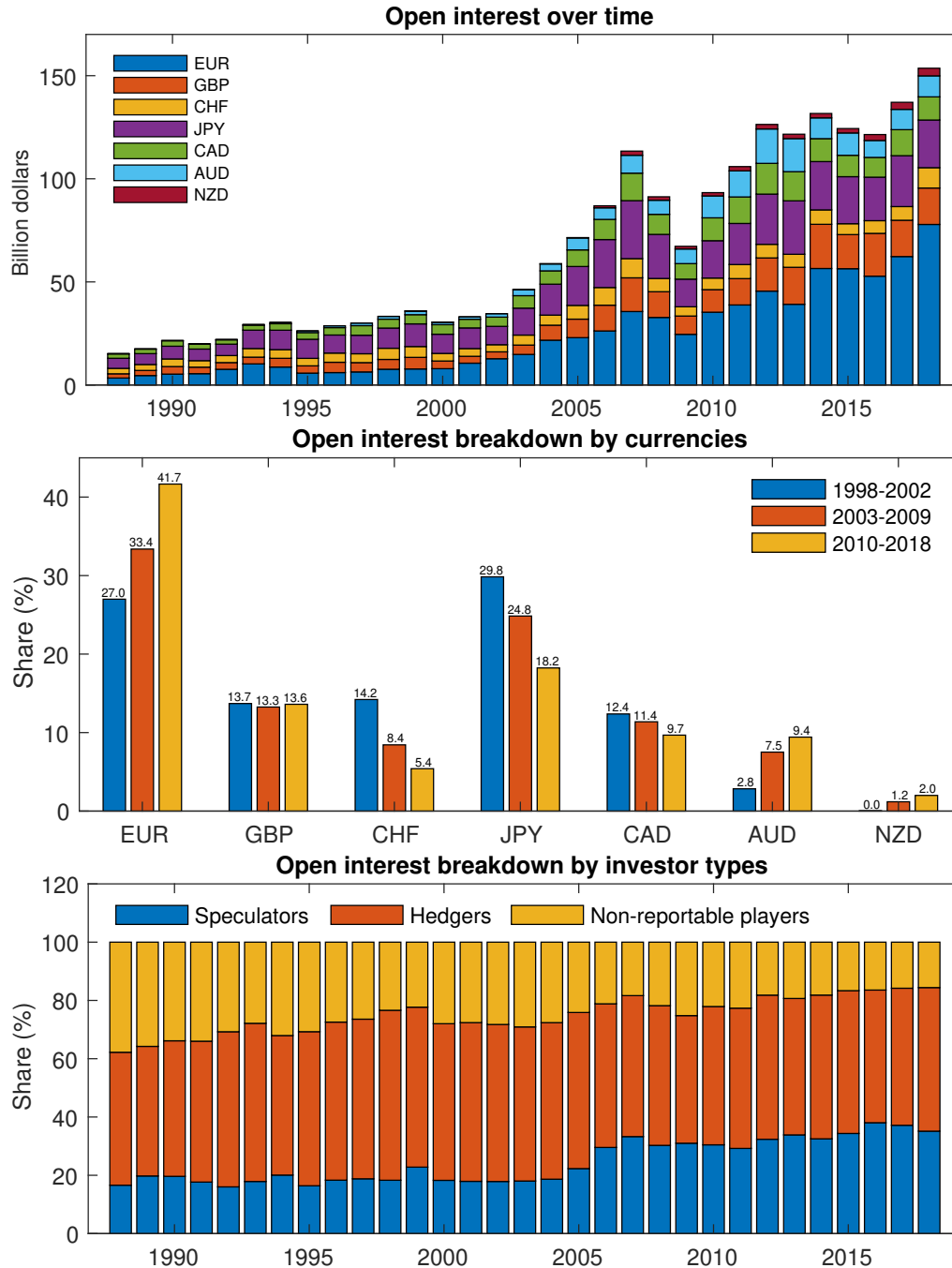


Figure 2: **Open interests in the FX futures market.** The top panel shows the open interest aggregate in U.S. dollars stacked over the seven largest FX futures. The middle panel breaks down the open interest aggregate by the selected futures. The bottom panel breaks down the open interest aggregate by different investor types.

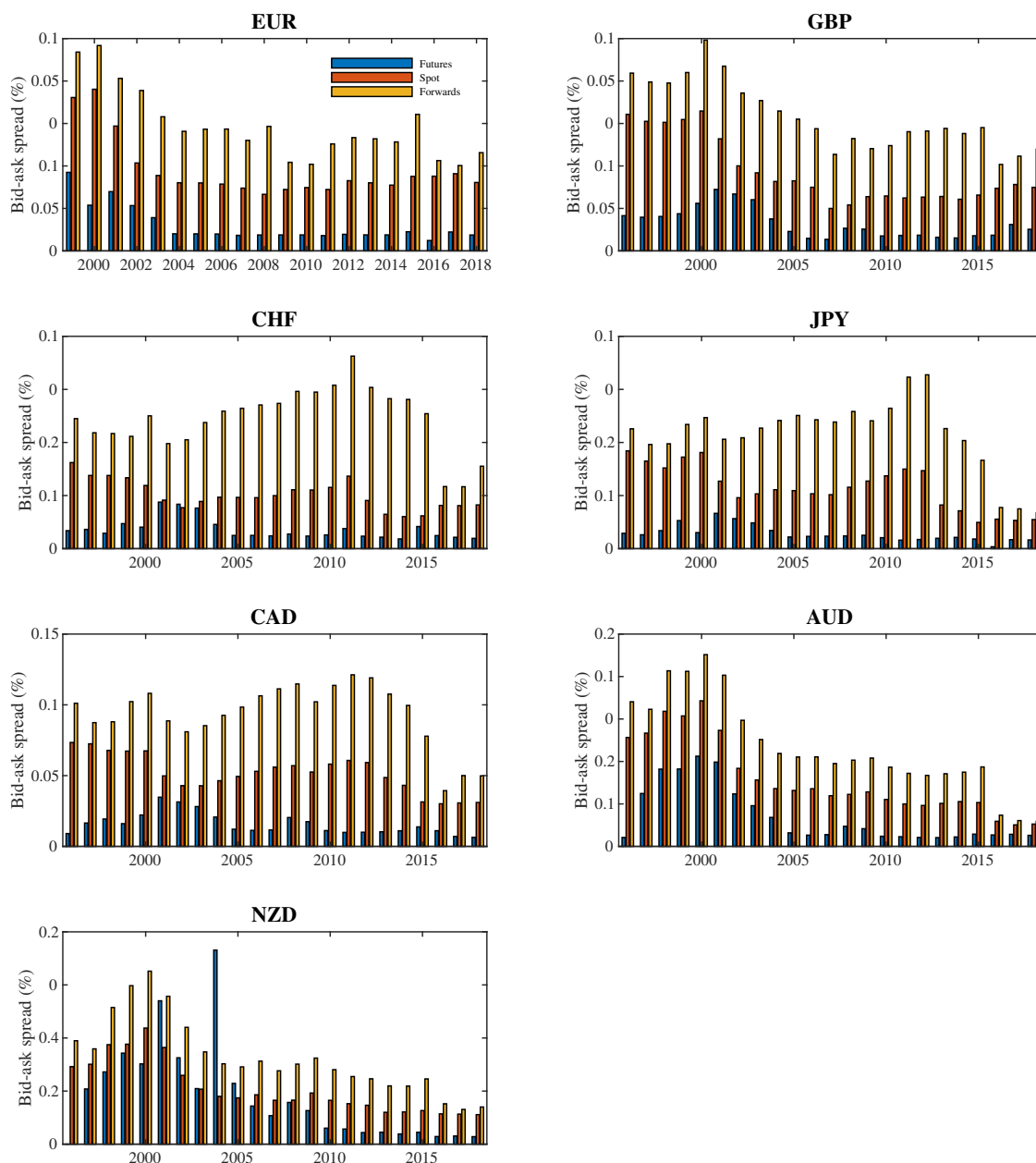


Figure 3: **Time series of bid-ask spreads on selected futures, compared with those on the OTC contracts.** The bid-ask spreads on futures are obtained by taking medians of bid-ask spreads during floor trading hours from 8:20 a.m. to 3:00 p.m. on the days of payroll announcements. The bid-ask spreads data on FX spot and three-month forwards are populated at the market close of payroll release days.

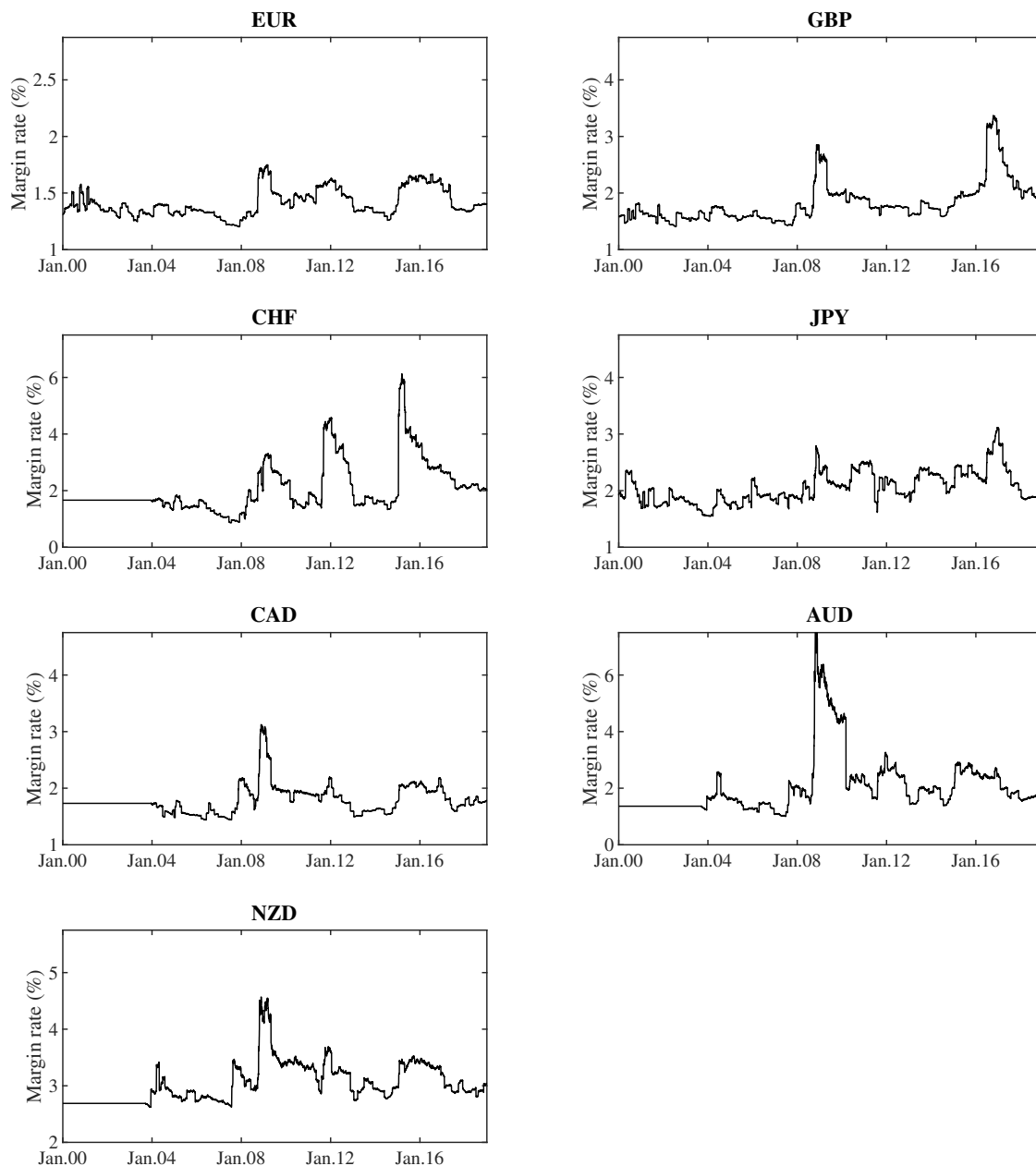


Figure 4: **Time series of individual overnight margin rates for selected futures.** As the CME margin is set in U.S. dollars per contract, I convert it into the margin rate—the margin divided by the futures price and by the contract size. The margin data start on different dates across contracts, so I fill the missing data backward for all contracts.

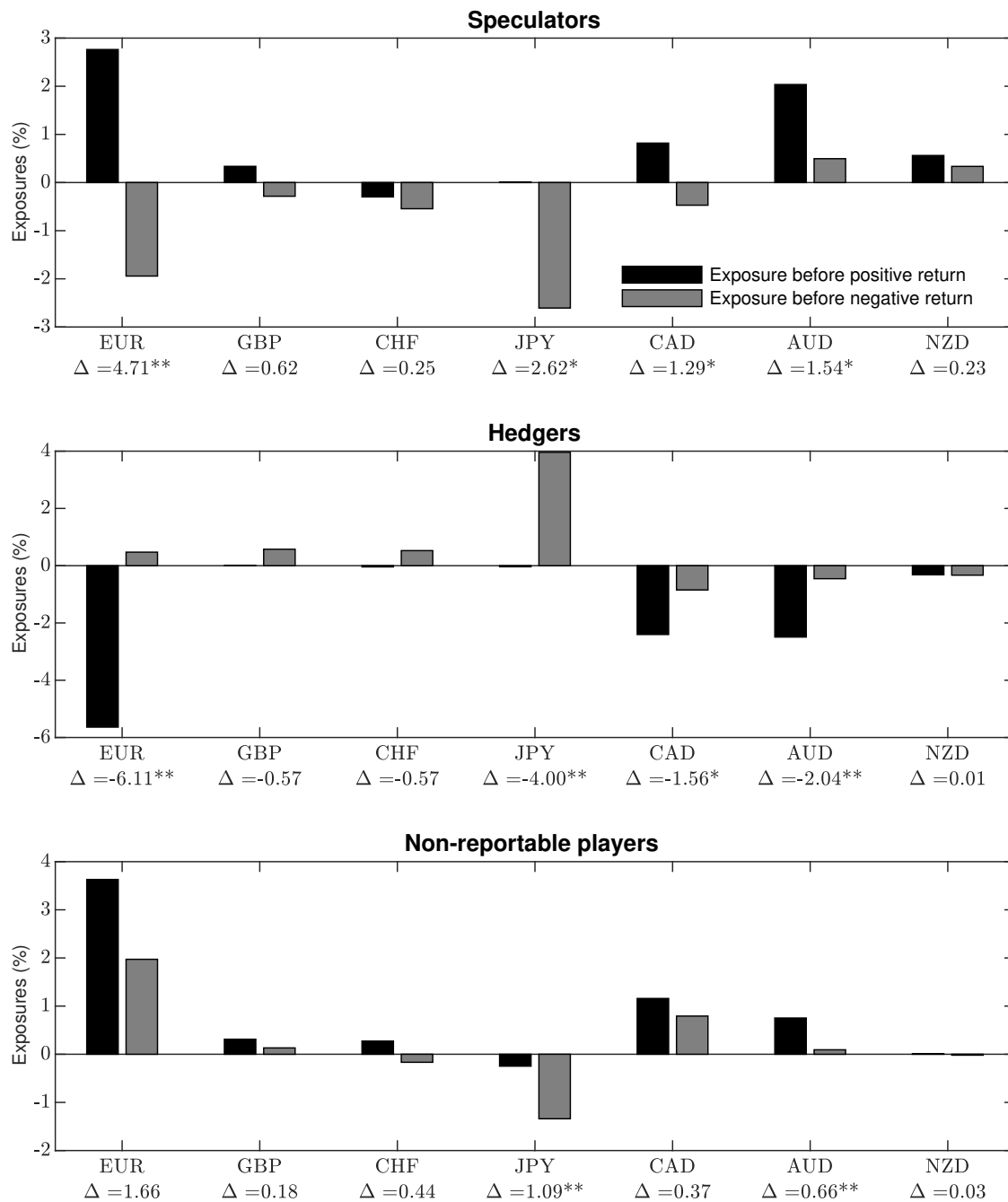


Figure 5: **Dirty FX exposures ahead of payroll announcements.** This figure compares average dirty FX exposures leading up to large positive payroll-news returns (the left bar for a given future) to those leading up to large negative payroll-news returns (the right bar for a given future). The figure shows the exposure difference between the two bars, denoted by Δ , for each future. The sample period covers January 1996 through December 2018. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively.

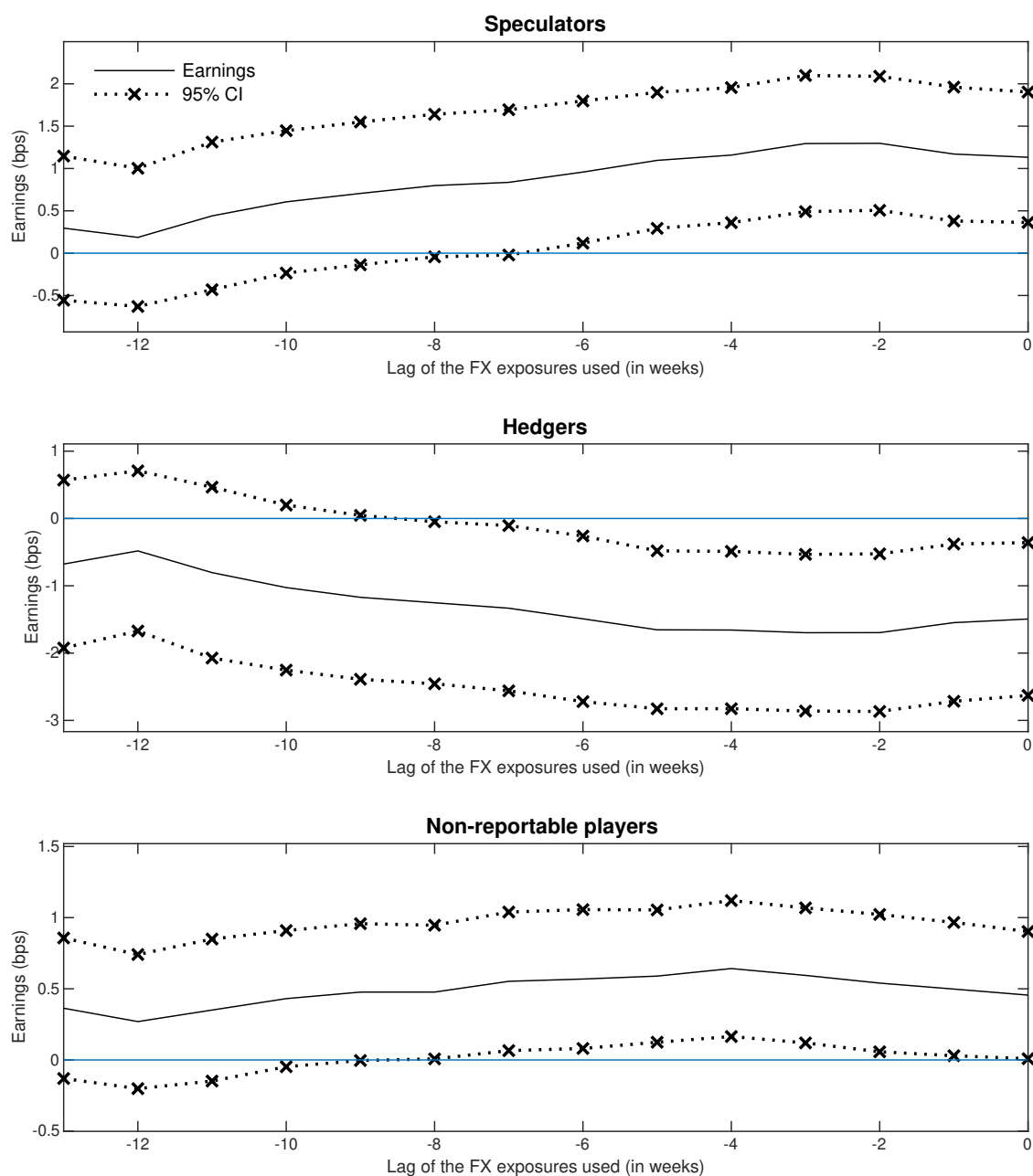


Figure 6: **Dirty portfolio earnings calculated using FX exposures with different lags.** The y axis represents the dirty earnings that each investor type might have earned on their entire FX portfolio around payroll announcements. The portfolio earnings are averaged over the sample period from January 1996 through December 2018. The x axis represents the lag (in weeks) of the underlying FX exposure data that are used to calculate the dirty portfolio earnings. Here, the intraday window is fixed at starting five minutes before announcements and ending 30 minutes after the announcements. The cross markers represent the 95% confidence interval.

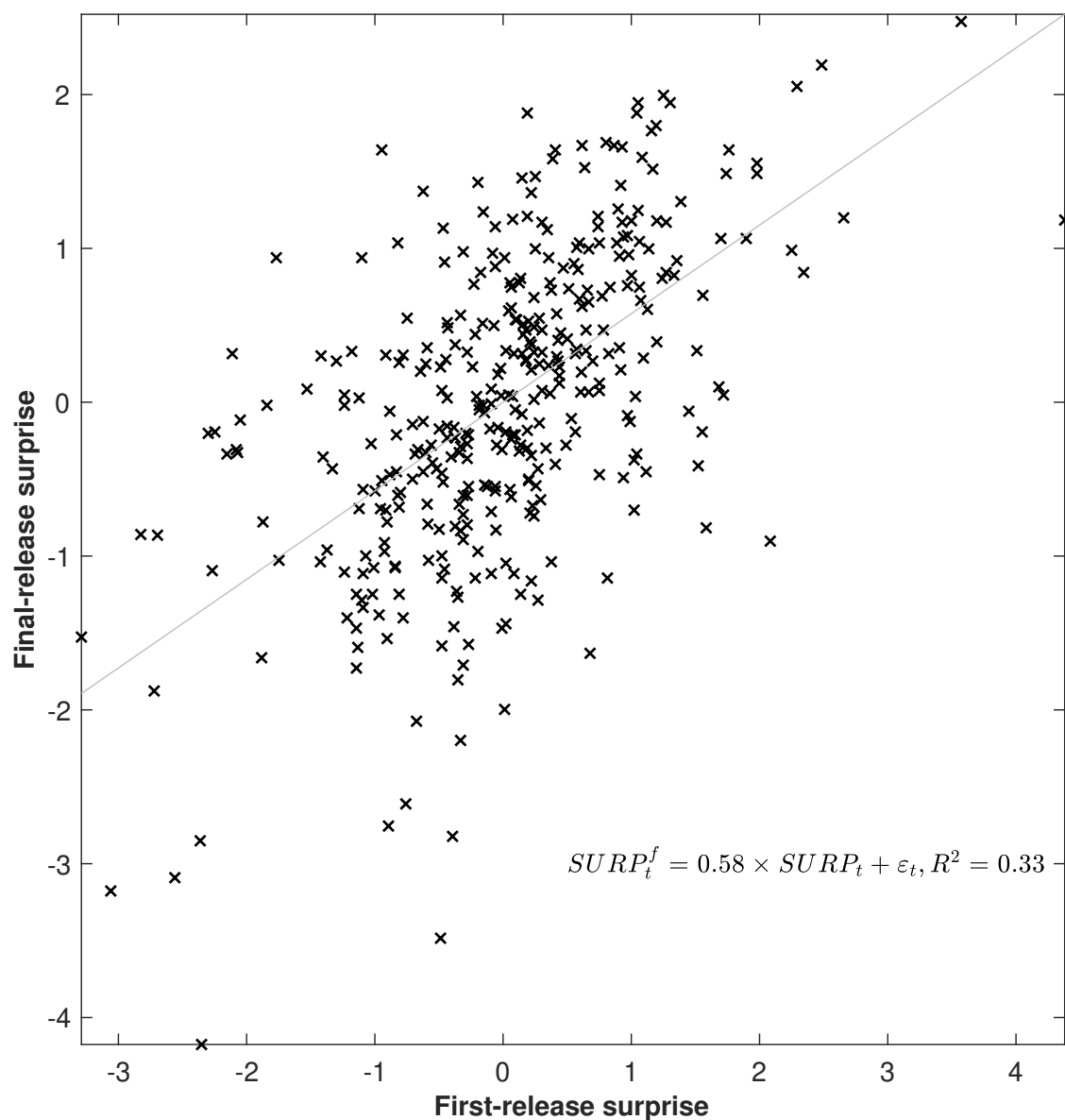


Figure 7: **Relationship between the first- and final-release payroll surprises.** $SURP_t$ denotes the first-release payroll surprise as defined in Equation 7. Similarly, I define the final-release payroll surprise ($SURP_t^f$) as the standardized difference between the finally revised payroll number and the consensus forecast. The sample spans from January 1988 to December 2018.

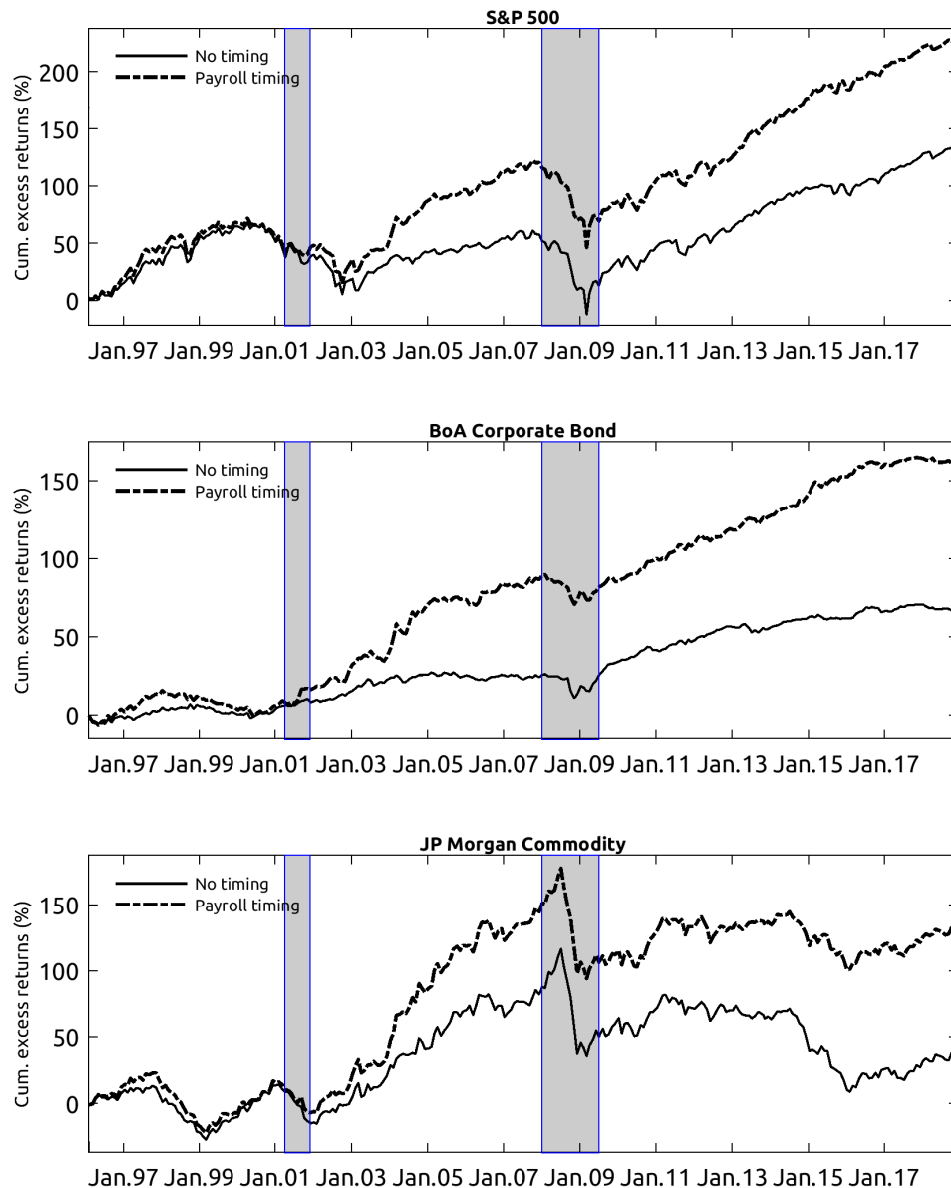


Figure 8: **Time series of cumulative excess returns of the payroll timing strategy in comparison to those with no timing strategy.** The payroll timing strategy assumes entering an FX portfolio that mimics speculators' clean exposures five minutes before a target payroll release and unwinding it one hour after that release. Three risky reference assets are considered: the S&P 500 total return index (the top panel), the Bank of America U.S. corporate bond total return index (the middle panel), and the JP Morgan aggregate commodity total return index (the bottom panel). Here, I assume 10% cash holdings, which are necessary for the required margin. The shaded areas refer to the NBER recessions included in the sample.

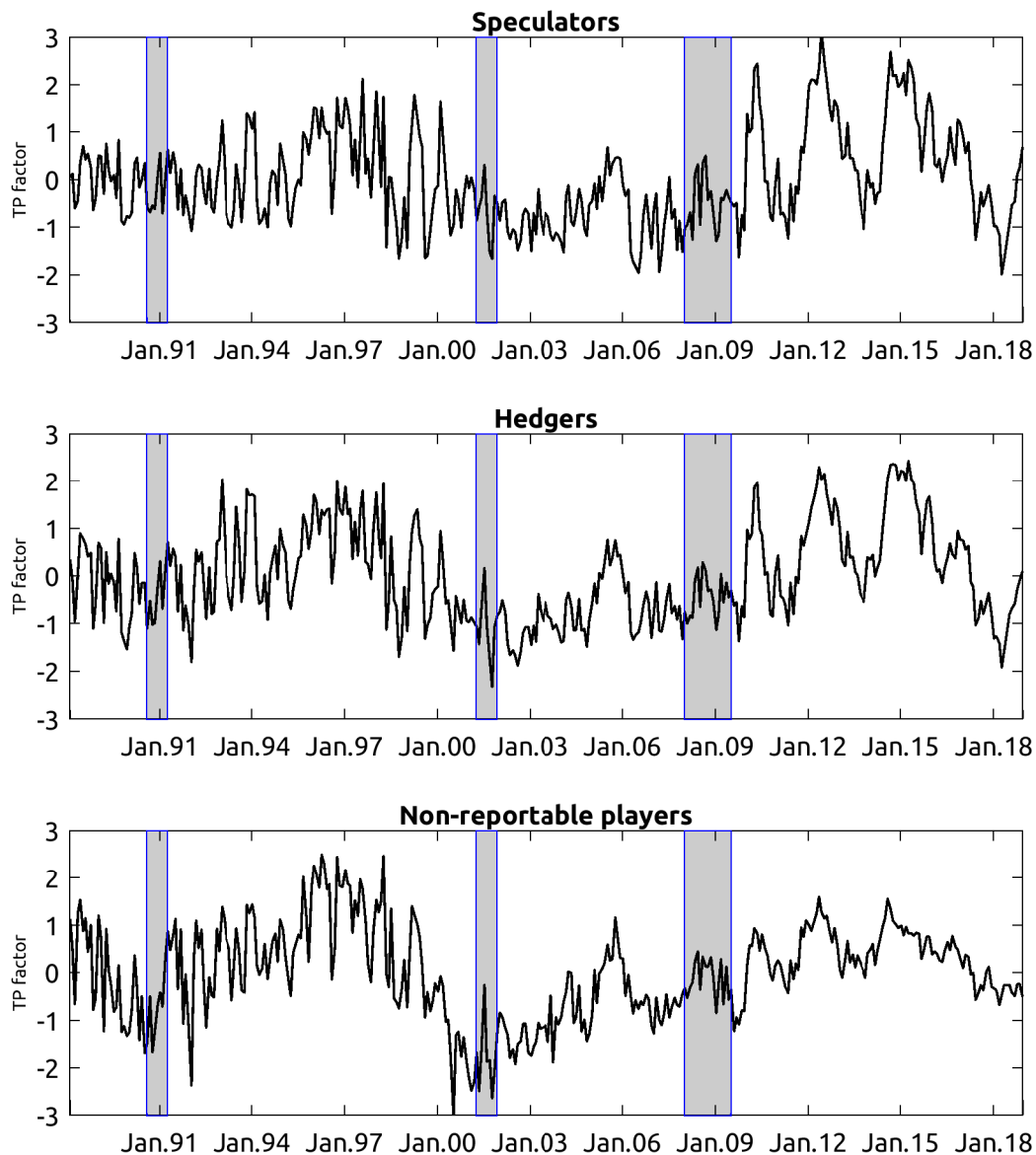


Figure 9: **Time series of the common factor of FX dirty exposures estimated via the Kelly and Pruitt (2015) three-pass filter.** The common factor is estimated such that the three-day-lagged dirty exposures are aligned with the impending payroll surprise. With that choice of factor proxy, the estimated factor is best positioned to capture the payroll information contained in FX exposures. Panels A, B, and C correspond to speculators, hedgers, and non-reportable players, respectively. The shaded areas refer to the NBER recessions included in the sample.

Appendix A Lead-lag relation among different investors'

FX exposures

Here, I provide additional evidence that speculators have acted like informed traders, whereas hedgers have provided liquidity. According to my finding, speculators (as informed traders) and hedgers (as liquidity providers) are likely to be active and passive investors, respectively. To compare eagerness to trade, I first extract a common factor of FX exposures (the first PCA component) for each investor type and then run Granger causality tests among three investor types' common factors. In particular, I expect that speculators' common factor might have led hedgers' common factor rather than vice versa.

Panel A of Table A.1 shows the Granger causality test results among different investor types' dirty common factors, which represent the maximum variations of dirty exposures. The cell (i, j) shows an F -statistic of the null that the row- i factor does not Granger-cause the column- j factor. The results are strikingly different between speculators and hedgers. Speculators' factor appears to Granger-cause that of hedgers at the 1% level (F -statistic=3.86) but not the other way around. Speculators' factor also appears to Granger-cause that of non-reportable players at the 10% level (F -statistic=2.47) but not the other way around. Interestingly, hedgers' factor is not even Granger-caused by its own lags (F -statistic=1.95).

Panel B of Table A.1 shows the Granger causality test results with different investor types' clean common factors, which represent the maximum variations of clean exposures. Speculators' factor appears to Granger-cause that of hedgers at the 1% level (F -statistic=7.60). In comparison, hedgers' factor appears to Granger-cause that of speculators at the 1% level (F -statistic=3.19). Although both directions are statistically significant, the former direction is stronger than the latter direction.

Overall, speculators' exposures appear to have driven hedgers' exposures rather than

vice versa, consistent with the main finding that speculators have acted like informed traders, whereas hedgers have provided liquidity.

Table A.1: Lead-lag relation among different investor types' FX exposures

This table shows Granger causality test results among different investor types' common exposures. The common exposure is obtained as the first PCA factor of dirty (in Panel A) or clean (in Panel B) exposures across different futures for each investor type. The optimal lag of a three-dimensional VAR model is determined by likelihood ratio tests at the 1% level for each panel. The cell (i, j) shows an F -statistic of the null that the row- i factor does not Granger-cause the column- j factor. ***, **, and * indicate statistical significance at the 1, 5, and 10% levels, respectively.

	Speculators	Hedgers	Non-reportable players
Panel A: Dirty common exposures			
Speculators	17.71***	3.86***	2.47*
Hedgers	0.66	1.95	1.10
Non-reportable players	1.26	1.71	28.42***
Panel B: Clean common exposures			
Speculators	9.96***	7.60***	8.24***
Hedgers	3.19**	6.09***	6.58***
Non-reportable players	3.10**	3.62**	33.47***

Appendix B Portfolio earnings around other macroeconomic announcements

I here examine speculators' dirty portfolio earnings around the FOMC and other macroeconomic announcements. With respect to the FOMC announcements, I use only the scheduled ones and account for the exact times that the announcements took place. To save space, I select six macroeconomic news considered to be most important in the literature, including real GDP advance, industrial production index, retail sales, consumer price index, producer price index, and personal consumption expenditures.

Figure A.1 presents speculators' dirty portfolio earnings around the FOMC and selected macroeconomic announcements. Here, the dirty portfolio earnings are calculated by applying Equation (5) to each of the FOMC and selected announcements. The figure shows no significant jump at 8:30 a.m. for every announcement type considered. In other words, I find little evidence of informed trading in FX futures with respect to the FOMC and macroeconomic announcements (other than payroll announcements).

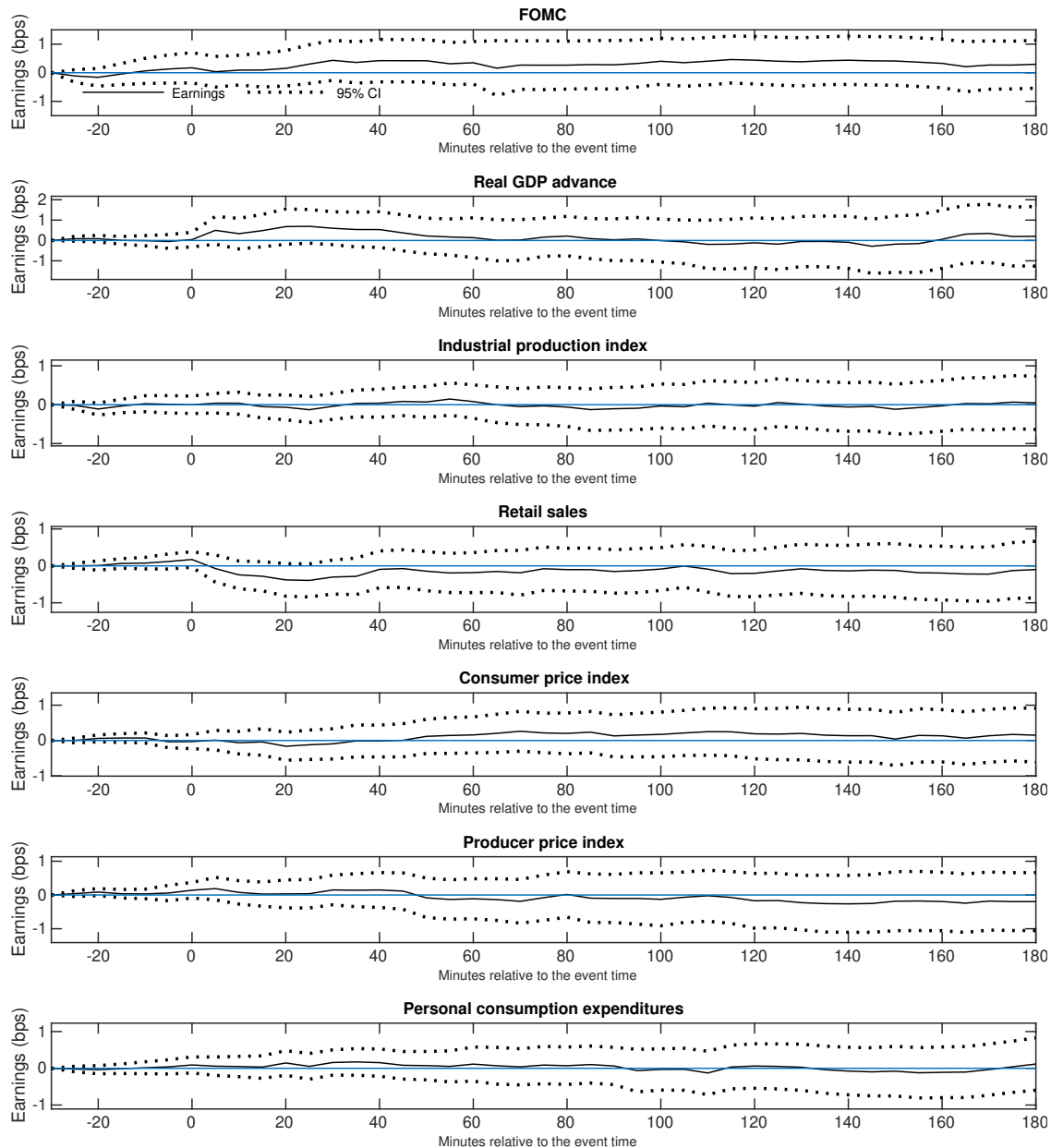


Figure A.1: **Speculators' dirty portfolio earnings around other macroeconomic announcements.** The y axis represents the dirty earnings that speculators might have earned on their entire FX portfolio around the FOMC and six selected macroeconomic announcements. The intraday window for computing the portfolio earnings starts 30 minutes before an announcement and lasts until three hours after the announcement. The portfolio earnings are averaged over the sample period from January 1996 to December 2018. The dotted line represents the 95% confidence interval.